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TOPICAL REVIEW

Photonic reservoir computing: a thematic review

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Abstract

Reservoir computing offers a versatile approach to temporal information processing by exploiting the dynamics of nonlinear systems to enable learning with a linear readout. Photonic hardware is particularly attractive for this paradigm, thanks to ultrafast dynamics and massive parallelism across multiple accessible degrees of freedom of light. In this thematic review, we chart the development of photonic reservoir computing across hardware platforms, from delay-based architectures and integrated photonic circuit implementations to free-space systems. We also discuss how the input and output layers influence the performance of photonic reservoirs. We present the explored learning algorithms that go beyond linear regression, and list some popular benchmark tasks and shed light on some key applications that have been investigated in the literature. Finally, we discuss emerging unconventional implementations. The overall goal of this review is to provide a panoramic overview of photonic reservoir computing, useful for both practitioners and newcomers to the field.

List of abbreviations

ADC	Analog-to-digital converter.
AI	Artificial intelligence.
AWG	Arbitrary waveform generator.
BER	Bit error rate.
BPTT	Backpropagation through time.
CA	Computational ability.
CCD	Charge-coupled device.
COVID-19	Coronavirus disease 2019.
CW	Continuous wave (source).
DAC	Digital-to-analog converter.
DDE	Delay differential equation.
DFB	Distributed feedback laser.
DMD	Digital micromirror device.
DPSK	Differential phase shift keying.
DSP	Digital signal processing.
EDFA	Erbium-doped fiber amplifier.
ELM	Extreme learning machine.

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ESN	Echo state network.
ESP	Echo state property.
FCD	Free-carrier dispersion.
FPGA	Field-programmable gate array.
GPU	Graphical processing unit.
GR	Generalization rank.
HAR	Human action recognition.
HOG	Histogram of oriented gradients.
i.i.d.	Independent and identically distributed.
InP	Indium phosphide.
IPC	Information processing capacity.
KQR	Kernel quality rank.
KS	Kuramoto–Sivashinsky.
LA-VCSEL	Large-area vertical-cavity surface-emitting laser.
LK	Lang–Kobayashi.
LSTM	Long short-term memory.
MAC	Multiply-accumulate.
MC	(linear) Memory capacity.
MFCCs	Mel-frequency cepstral coefficients.
MMF	Multimode fiber.
MNIST	Modified National Institute of Standards and Technology (database).
MRR	Microring resonator.
MZI	Mach–Zehnder interferometer.
MZM	Mach–Zehnder modulator.
NARMA	Nonlinear autoregressive moving-average (time series).
NG-RC	Next-generation reservoir computing.
NMSE	Normalized mean square error.
NN	Neural network.
NVAR	Nonlinear vector autoregression.
PAM	Phase amplitude modulation.
PCA	Principal component analysis.
PCM	Phase-change material.
PD	Photodiode.
PIC	Photonic integrated circuit.
PLC	Planar lightwave circuit.
PNN	Photonic neural network.
PUF	Physical unclonable function.
QAM	Quadrature amplitude modulation.
RC	Reservoir computing.
ReLU	Rectified linear unit.
RF	Radio frequency.
RNG	Random number generator.
RNN	Recurrent neural network.
SER	Symbol error rate.
SiN	Silicon nitride.
SiPh	Silicon photonics.
SL	Semiconductor laser.

SLM	Spatial light modulator.
SNN	Spiking neural network.
SNR	Signal-to-noise ratio.
SOA	Semiconductor optical amplifier.
TDRC	Time-delay reservoir computing.
TFLN	Thin-film lithium niobate.
TIA	Transimpedance amplifier.
TPA	Two-photon absorption.
VCSEL	Vertical-cavity surface-emitting laser.
VLSAI	Very large scale artificial intelligence.
VMM	Vector-matrix multiplication.
WDM	Wavelength-division multiplexing.
WER	Word error rate.

1. Introduction

Most current digital hardware remains rooted in von Neumann flavored architectures, where memory and processing units are physically separated and communicate via electrical interconnects. This arrangement results in the majority of the energy being dissipated as heat during data transfer. Thus, a growing share of the energy budget is tied to moving data and cooling the generated heat rather than actual processing, which motivates shifting communication into the photonic domain to reduce such costs [1–4]. By exploiting light for data transfer, photonic systems can substantially reduce thermal losses and minimize the need for extensive physical interconnections, particularly through the use of WDM techniques. Today, optical links are already mitigating interconnect bottlenecks at rack and system scales [3], and continued integration—up to and including co-packaged optics—is reshaping short-reach networks, according to recent industry reports [5–7].

Meanwhile, unconventional computing paradigms are attracting intense interest, driven by escalating demands on bandwidth and energy efficiency, particularly in the context of modern data centers. These demands manifest in billions of computationally intensive operations being performed everyday on inefficient digital hardware. Within this context, photonics is explored both for acceleration (e.g. optical matrix multipliers and tensor cores [8, 9]) and for unconventional computing paradigms where the physics of light is naturally advantageous [10–12]. On the one hand, emerging photonic accelerators [13], recently developed through advances in photonic integration, have the advantage of fundamental compatibility with existing computing architectures, which brings them closer to practical fabrication and industrial adoption. Photonic accelerators for VMM/MAC exploit the fact that light can passively perform linear transformations, i.e. without an additional external energy source. Such architectures are realized in integrated photonics, such as interferometric meshes and microring arrays, and in free-space setups such as diffraction-based systems. On the other hand, photonic hardware is also emerging as a promising platform for unconventional computing paradigms, driven by its THz-scale bandwidths and potential for superior energy efficiency [14], especially in the context of neuromorphic computing [15–17].

RC is an unconventional computing paradigm that has been independently formulated in the foundational works of Jaeger [18] and Maass *et al* [19]. RC harnesses a nonlinear dynamical system for information processing. In standard RC, an input drives the dynamical system, which it then transforms deterministically according to some nonlinear function, and then outputs a signal in which the input features are embedded. A simple, often regularized, regression can be used to find the optimal output weights to perform the task at hand [20]. Key to the use of such simple training methods, is the nonlinear transformation in the reservoir (a network of nonlinear nodes), which provides a so-called dimensionality expansion which enables a larger overall 'separation' of input features. The weights on the connections from the input layer to the reservoir layer, and the intra-reservoir connections, are fixed in principle. However, in practical implementations, a handful of hyperparameters related to these weights are also usually adjusted and optimized to set the reservoir to the desired dynamical regime where it performs the task at hand with sufficient accuracy. In some cases but not always, the reservoir hyperparameters are adjusted in such a way as to bias the dynamical system at a point near the 'edge of stability', e.g. close to a bifurcation point, to yield a sufficiently rich response and maximize the retention of the

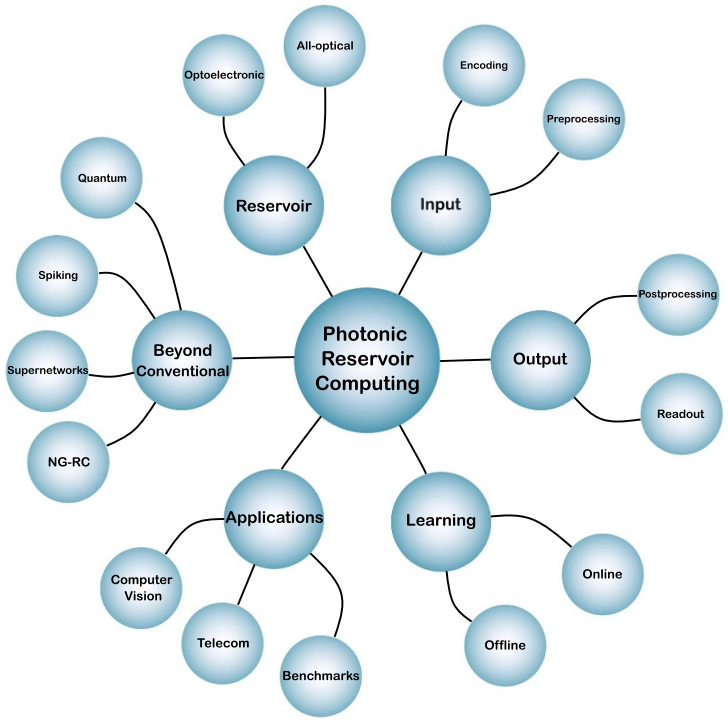


Figure 1. Photonic reservoir computing divided into key themes and subtopics, as discussed throughout the paper.

reservoir [21]. Over the years, RC has demonstrated competitive performance across tasks such as channel equalization, spoken digit recognition, and chaotic time series forecasting, among others, on a wide range of platforms [22], and especially in photonics [23].

The multitude of implementations and the rising popularity of photonic RC over the years, in addition to the sprawling nature of the relevant literature, call for a cohesive review which summarizes the main directions explored in photonic RC. Over the years, photonic RC has been the subject of multiple reviews [23, 24], perspectives [25, 26], and detailed tutorials [27, 28]. We build on the existing works by systematically reporting the overall advances and trends in photonic RC since its inception and up to the latest advances at the time of writing, while simultaneously identifying common threads across the literature.

We have thus chosen a structure based on the themes that have been explored in the photonic RC literature thus far, as graphically illustrated in figure 1. In section 2, we give an introduction to RC theory and present an overview of the hardware implementations proposed and demonstrated as photonic reservoirs. In sections 3 and 4, we report the hardware implementations of the input and output layers, respectively, as well as the explored preprocessing and postprocessing techniques, respectively. In section 5, we report the various methods explored for hyperparameter optimization and training. In section 6, a comprehensive list of the most widely used evaluation methods and benchmarks is given, as well as real-world applications that have been explored thus far. In section 7, we report the works which fall under the broad category of unconventional photonic RC. Finally, we present a summary and outlook for the field of photonic RC in section 8.

2. Photonic reservoirs

2.1. Overview of RC

The ESN [18] is a paradigmatic example of the RC framework, which leverages a nonlinear dynamical system with short term memory, exhibiting two fundamental properties:

- ESP: closely related to the *fading memory* [29], ensures that the influence of past inputs on the reservoir state diminishes over time, allowing the system to forget its initial conditions and produce consistent responses to similar inputs.

- Dimensionality expansion: typically realized through a nonlinear activation function which maps the input data into a higher dimensional space, thereby improving the linear separability of input patterns.

The ESP is essential for ensuring that the reservoir's behavior is consistent, such that its state becomes dependent only on the recent input history and not on its initialization history. This is crucial to obtain a trainable reservoir response, where the system produces consistent internal representations for identical inputs. Dimensionality expansion enables complex input features to be well-separated by projecting the input onto a higher dimensional space. This is where the computing takes place. The well-separated mapping can then be 'learned' using a simple model, such as linear regression. In general, any RC scheme is composed of three main layers:

- Input layer: where the input signal is injected into the reservoir;
- Reservoir layer: a network of nonlinear, recurrently connected nodes;
- Output layer: records the reservoir states and generates the system's output through a (typically linear) readout function.

In contrast to standard RNNs where all the layers must be trained, in RC the weights of the input and reservoir layers are, in principle, fixed to arbitrary values. Only the output weights need to be trained. In an ESN, the discrete time evolution of all nodes in the reservoir generally follows:

$$\mathbf{x}[n] = f_{\text{res}}\left[\mathbf{W}^{\text{in}}\,\mathbf{u}[n] + \mathbf{W}^{\text{int}}\,\mathbf{x}[n-1]\right], \tag{1}$$

where f_{res} is the reservoir's (nonlinear) activation function, $\mathbf{x}[n] \in \mathbb{R}^N$ is the response of the N reservoir nodes at discrete time step n, $\mathbf{u}[n] \in \mathbb{R}^M$ is the input at n (with typically $M = 1$), $\mathbf{W}^{\text{in}} \in \mathbb{R}^{N\times M}$ is the input-to-reservoir weight matrix, and $\mathbf{W}^{\text{int}} \in \mathbb{R}^{N\times N}$ is the recurrent weight matrix that defines the internal reservoir connectivity. The prediction is computed through a readout of the reservoir state:

$$\hat{\mathbf{Y}} = f_{\text{out}}\left(\mathbf{X}\mathbf{W}^{\text{out}}\right), \tag{2}$$

where $\hat{\mathbf{Y}} \in \mathbb{R}^{L\times S}$ is the matrix of predictions, L is the number of training examples, f_{out} is the output function (typically linear in standard software-based ESNs, but not necessarily so in practical implementations as will be discussed later), $\mathbf{W}^{\text{out}} \in \mathbb{R}^{N\times S}$ is the output weight matrix, where S is the output target dimensionality, and $\mathbf{X} \in \mathbb{R}^{L\times N}$ is the concatenated state matrix. If a bias term is included, the network will have one additional unconnected node, thus $\mathbf{X} \in \mathbb{R}^{L\times(N+1)}$ and $\mathbf{W}^{\text{out}} \in \mathbb{R}^{(N+1)\times S}$. In classical ESNs employing a tanh activation function, the ESP is satisfied—for sufficiently large reservoirs [30]—by scaling the spectral radius of the internal weights matrix such that $\rho\left(\mathbf{W}^{\text{int}}\right) = \max_i |\lambda_i| < 1$. The weights are found using the Moore–Penrose pseudoinverse:

$$\mathbf{W}^{\text{out}} = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{Y}, \tag{3}$$

where $\mathbf{Y} \in \mathbb{R}^{L\times S}$ is the matrix of targets (labels). Sometimes, but not always, Tikhonov regularization is added to mitigate overfitting:

$$\mathbf{W}^{\text{out}} = \left(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}\right)^{-1}\mathbf{X}^\top\mathbf{Y}, \tag{4}$$

where λ is the Tikhonov regularization parameter and $\mathbf{I}$ is the identity matrix. This is also known as ridge regression.

In physical RC, the hardware—or physical substrate—serves as the reservoir, with its underlying physics realizing the essential reservoir properties discussed above. A physical reservoir typically realizes the ESP, or fading memory, through dissipative dynamics of its state variable(s) around a unique stable attractor. In photonic RC, this is typically achieved through an optical cavity or a hybrid optoelectronic cavity. A nonlinearity is typically implemented using either nonlinear optical phenomena, or through exploiting a nonlinear transformation of the state variable(s) during the input injection or readout processes, or both. The effective dimensionality of a photonic reservoir's phase space is determined by the dimensionality of its intrinsic state variable(s), together with how the dynamical regime is tuned to unfold these states across time, space, frequency, or other available physical degrees of freedom in the optical system. This wide range of accessible dimensions is what enables massive parallelism in photonic RC. Regarding the input and readout, although photonic reservoirs *compute* in continuous

time, they are typically driven and observed in a discretized manner. The input is usually injected into the continuous-time system via sample-and-hold, while the continuous time output is then sampled to obtain discrete states.

Motivated by the advantages of photonic hardware for computation [14], photonic RC has already been demonstrated in a wide variety of implementations. Beyond supplying the required nonlinearity and feedback, practical designs must also balance power consumption, footprint, and hardware feasibility (including control and calibration of setups). This section discusses (i) photonic TDRC, (ii) spatially distributed photonic RC, and (iii) common experimental challenges and how they have been addressed in the literature.

2.2. Time-delay photonic RC

TDRC and its foundations were introduced in [31, 32]. There are subtleties to working with delay dynamical systems as reservoirs, which we briefly discuss. Input-driven, single-delay dynamical systems with P independent, continuously evolving variables, can be described by a DDE of the general form:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}(t) = F(\mathbf{x}(t), \mathbf{x}(t-\tau_{\mathrm{fb}}), \mathbf{u}(t)), \tag{5}$$

where $F : \mathbb{R}^P \times \mathbb{R}^P \times \mathbb{R}^M \to \mathbb{R}^P$ is a function of the present state, the delayed state, and the external input, $\tau_{\mathrm{fb}} > 0$ denotes the feedback delay, $\mathbf{x}(t) \in \mathbb{R}^P$ is the state vector, and $\mathbf{u}(t) \in \mathbb{R}^M$ is the external input signal.

Unlike ordinary differential equations, the state of a DDE at time t depends not only on the instantaneous value $\mathbf{x}(t)$ but on the entire history segment $\mathbf{x}(s)$, $s \in [t-\tau_{\mathrm{fb}}, t)$ [33]. As a consequence, the system's phase space is formally infinitely dimensional. In the context of RC, however, one does not exploit the full infinite-dimensional state. Instead, the delay interval is discretized into N taps (so-called 'virtual' nodes), yielding a finite-dimensional approximation of the underlying delay dynamics. Thus, in principle, the dimensionality is unbounded, but in practice the effective reservoir size is fundamentally limited by, for example, bandwidth and noise, among other limitations. As a rough example, an input stream of length $L = 1000$ injected into a TDRC with $\tau_{\mathrm{fb}} = 10\,\mathrm{ns}$ and sampled at a period of 1 ns, will yield a reservoir size $N = 10$ and a 1000×10 state matrix (assuming no bias term). A critical input preprocessing protocol called the mask is typically applied, which will be discussed later in section 3.

Another subtlety is that the topology of the reservoir, which defines the connectivity of the (virtual) nodes, can be understood by considering the interplay between the input clock cycle T_{clk}, τ_{fb}, the temporal distance between adjacent nodes θ (equivalent to the masked input injection period), and the characteristic timescale of the nonlinear node t_{nl} [34]. First, consider the case when $t_{\mathrm{nl}} > \theta$ (figure 2(c)). This means that the masked input would be injected during the transience of the response to the previous masked input(s). In this sense, each node would forward-couple to the $\lfloor t_{\mathrm{nl}}/\theta \rfloor$ following nodes. Second, consider the case when $t_{\mathrm{nl}} \ll \theta$, and the synchronous operation, i.e. setting $T_{\mathrm{clk}} = \tau_{\mathrm{fb}}$. In this case, the nodes are only receiving their own feedback governed by t_{nl} (essentially there is no 'network', as shown in figure 2(d). As introduced in [35], offsetting τ_{fb} and T_{clk} allows the forward coupling of nodes despite t_{nl} being too small. This is done by setting $\tau_{\mathrm{fb}} = T_{\mathrm{clk}} + \theta$, yielding a cyclic topology where each node is forward-coupled exactly to the following one (figure 2(e)). While insightful, these earlier notions mostly emanated from viewing time delay reservoirs in terms of their network equivalents.

In the more recent literature, it was shown that the choice of T_{clk} relative to τ_{fb} can be made with greater flexibility [36], with one notable exception: the case of resonance between the two. The synchronous operation in TDRC was found to degrade the performance on the linear MC and several benchmark tasks in multiple studies [37, 38]. In general, the performance was found to deteriorate when the resonant case is satisfied between T_{clk} and τ_{fb}, i.e. when $aT_{\mathrm{clk}} = b\tau_{\mathrm{fb}}$, where $a, b \in \mathbb{N}$. According to the authors, the rationale behind this degradation in MC is that the equivalent networks in the resonance regime will have their information overwritten very quickly by new inputs. It is likely that this degradation would harm, by extension, the IPC [39] of TDRC schemes. On the other hand, the best performance is achievable in regions of non-resonant clock cycles.

2.2.1. Optoelectronic implementations

The first experimental demonstrations of photonic RC [35, 40] employed a MZM in a feedback configuration to realize a TDRC scheme. As shown in figure 3(a), the optoelectronic feedback consists of an optical delay line, followed by a photodetector which converts the states back to the electrical domain, to be combined with the input data, and injected into the MZM, which modulates the CW laser source according to the RF input. Assuming unbalanced operation, the output field $E_{\mathrm{mzm}}(t)$ is proportional to

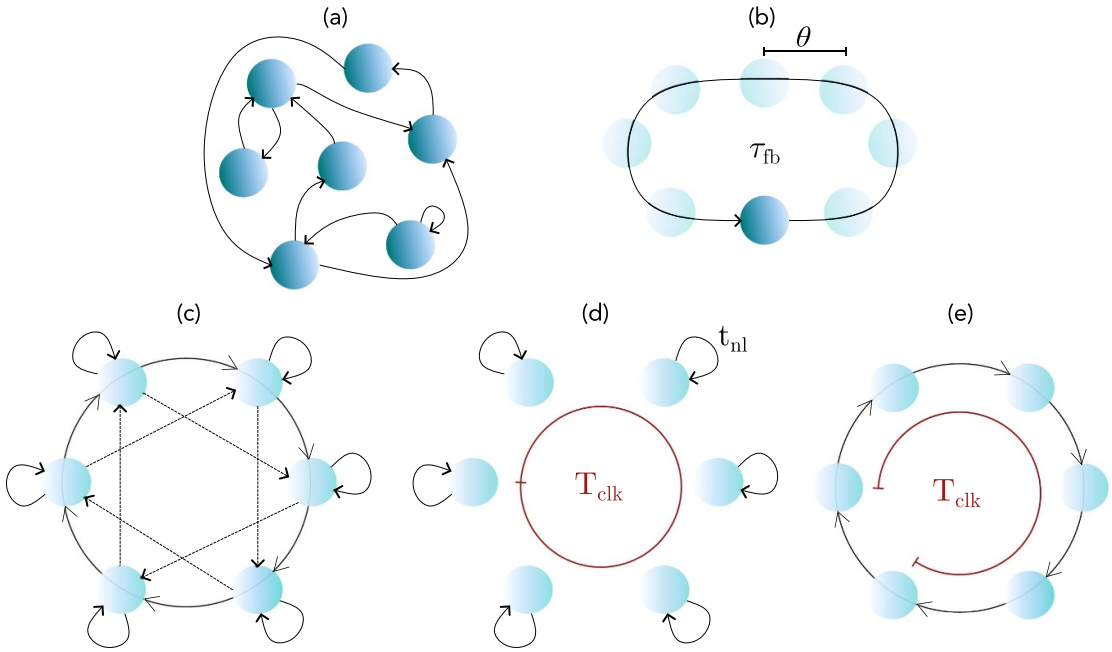


Figure 2. Physical reservoirs can be implemented either as (a) a spatial network of interconnected nonlinear elements (nodes) or (b) a single nonlinear element with feedback, where virtual nodes are created through time-multiplexing. Depending on the interplay of the relevant timescales (see section 2.2), the temporal connectivity in TDRC can manifest in different configurations (c)–(e).

the voltage applied $V(t)$ (including the DC offset) to modulate the input field E_{in}:

$$E_{\text{mzm}}(t) = E_{\text{in}} \cos\left(\frac{\pi V(t)}{V_\pi}\right), \tag{6}$$

where V_π denotes the zero intensity bias point. This optoelectronic system can be described by the well known Ikeda dynamics [41]:

$$\epsilon \frac{\mathrm{d}}{\mathrm{d}t} x(t) = -x(t) + \beta \sin^2(x(t-\tau_{\text{fb}}) + \gamma \mathbf{u}(t) + b), \tag{7}$$

where β is the feedback strength, τ_{fb} is the feedback delay time, $\mathbf{u}$ is the input, γ is the input scaling factor, b is the bias, and ϵ scales the response rate of the system. The $\sin^2(\cdot)$ dependence originates from the power transfer function of the modulator. In [40], the Ikeda system was employed for photonic RC to realize 400 time-multiplexed nodes, and was operated at 50 kHz (corresponding to a delay time of $\tau_{\text{fb}} = 20.9\,\mu$s) and was used to successfully solve the Santa Fe one-step-ahead prediction task and the spoken digit recognition task (benchmark tasks are discussed in detail in section 6). In [35], the authors employed the same reservoir to realize 50 nodes driven at approximately 118 kHz ($\tau_{\text{fb}} = 8.5\,\mu$s). The system's performance on a variety of benchmark tasks was reported, including the NARMA-10 task, a signal classification task, nonlinear channel equalization, and isolated spoken digit recognition. Besides the optoelectronic Ikeda system, a few other schemes have been investigated for optoelectronic RC. In [42], an electro-optic phase modulation of a CW laser is exploited to provide the nonlinearity. A DPSK Mach–Zehnder demodulator provides nonlocal connectivity, where the time-multiplexed nodes are coupled to several nearby nodes. The scheme was investigated numerically and tested experimentally on the speech recognition task, achieving processing speeds of one million words per second with low classification error. Optoelectronic reservoirs based on SLs, shown in figure 3(b), have been theoretically investigated [43], where their MC and CA were quantified, and recently demonstrated in experiment [44], reporting good performance on the Mackey–Glass time series. SL-based reservoirs are usually implemented with optical feedback, as one kind of all-optical approaches, which are discussed in the following section.

2.2.2. All-optical implementations

All-optical RC refers to schemes in which the reservoir's internal dynamics and recurrent feedback are implemented entirely in the optical domain, including the input injection and optical feedback paths.

Auxiliary electronics are typically used at the periphery-for driving modulators and for photodetection and post-processing-but the reservoir itself remains optical. Consequently, the potentially accessible processing bandwidth is higher than in optoelectronic implementations, although practical limits are set by modulator/detector bandwidths and the response time of the exploited nonlinearity.

The first reported all-optical demonstration utilized the gain saturation nonlinearity of a SOA [45]. The shape of the nonlinearity, i.e. the power transfer function, could be adjusted based on the injection current. This implementation allowed to solve the channel equalization task, the IPIX radar task, and the speech recognition task. Shortly afterwards, an implementation using a SL in optical feedback [46], depicted in figure 3(c), was reported. SLs were later extensively used in the photonic RC literature. The consistency of their dynamics was studied in [47, 48], and recently revisited in a high resolution parametric study in [49]. The dynamics of the system can be modeled in most cases by the well-known LK equations [50], extended to include the optical injection term:

$$\frac{\mathrm{d}}{\mathrm{d}t}E_{\mathrm{cav}}(t) = \frac{1}{2}(1+\mathrm{i}\alpha)\left[G(E_{\mathrm{cav}},N_{\mathrm{c}}) - \frac{1}{t_{\mathrm{p}}}\right]E_{\mathrm{cav}}(t) + \kappa_{\mathrm{fb}}\mathrm{e}^{-\mathrm{i}\phi}E_{\mathrm{cav}}(t-\tau_{\mathrm{fb}}) + \kappa_{\mathrm{in}}E_{\mathrm{in}}(t)\,\mathrm{e}^{-\mathrm{i}\Delta\omega t}, \tag{8}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}N_{\mathrm{c}}(t) = J - \frac{1}{t_{\mathrm{e}}}N_{\mathrm{c}}(t) - G(E_{\mathrm{cav}},N_{\mathrm{c}})\,|E_{\mathrm{cav}}(t)|^2, \tag{9}$$

where $E_{\mathrm{cav}}(t)$ denotes the (complex) intracavity electric field, and $N_{\mathrm{c}}(t)$ represents the carrier density in the gain medium, α is the linewidth enhancement factor, t_{p} is the photon lifetime, κ_{fb} is the feedback strength, ω_0 is the optical angular frequency of the laser, and ϕ is the feedback phase. The term $\kappa_{\mathrm{in}}E_{\mathrm{in}}(t)\mathrm{e}^{-\mathrm{i}\Delta\omega t}$ accounts for the injection of an external optical field into the laser cavity. In this expression, κ_{in} is the injection strength, and the term $E_{\mathrm{in}}(t)$ represents the injected field; in this case it is time-varying and encoded using, for example, an MZM. The exponential factor $\mathrm{e}^{-\mathrm{i}\Delta\omega t}$ accounts for the frequency detuning between the injected field and the field inside the SL cavity, where $\Delta\omega = \omega_{\mathrm{in}} - \omega_0$. The input injection term models the coupling of the externally injected optical field into the cavity, allowing the encoded input signal to directly modulate the laser dynamics and, depending on strength and detuning, potentially driving the system toward injection-locked operation or into other nonlinear regimes such as bistability or chaos [51]. It is worth noting that equations (8) and (9) assume only parallel polarization. For the carrier rate equation, the injection current density is denoted by J, q is the elementary charge, the carrier lifetime is t_e, and $G(E_{\mathrm{cav}},N_{\mathrm{c}})$ is the gain term, which depends on both the carrier density and the optical intensity. The gain can be modeled as a saturable function:

$$G(E_{\mathrm{cav}},N_{\mathrm{c}}) = g\frac{N_{\mathrm{c}}(t)-N_0}{1+\eta|E_{\mathrm{cav}}(t)|^2}, \tag{10}$$

where g is the differential gain coefficient, N_0 is the transparency carrier density, and η is the gain saturation coefficient. The denominator provides a macroscopic description of nonlinear gain saturation at high optical intensities, which originates from spectral hole burning. Typical values for SL parameters can be found in [52].

In a pioneering study [46], the SL-based reservoir was driven at 1 Gbps and was tested on speech recognition and the Santa Fe time series It was tested under both electrical and optical injection, and showed that the best results were obtained using a polarization-rotated feedback configuration. It was then augmented by a detailed numerical study [53], which showed the performance capabilities of SL-based reservoirs under electrical or optical injection, and the effect of the feedback polarization. Electrical injection was further investigated in [54], where state of the art results were obtained on the Santa Fe and nonlinear channel equalization tasks by careful optimization of τ_{fb} and κ_{fb}, and it was shown that increasing J can speed up the overall processing speed in this context. The role of phase dynamics was explored numerically in [55]. Specifically, this study highlighted the much faster relaxation frequency of the phase dynamics compared to the relaxation of the intensity dynamics under certain operating conditions, enabling faster input injection rates. In the same direction, a numerical [56] (experimental [57]) study showed the possibility of further speeding up the system, reaching a small $\theta = 12$ ps using a higher input optical intensity and carefully choosing $\Delta\omega$. This was built upon in the following works [58, 59], which utilized the enhanced bandwidth regime in addition to the asynchronous regime. In those works, T_{clk} is set much smaller than τ_{fb}, while showing performance improvement on the Mackey–Glass time series, compared to the synchronized case. Similarly, in [60], T_{clk} was smaller than τ_{fb}, and good performance was obtained on the Santa Fe task. This desynchronization between input and output was also utilized in other all-optical implementations [61]. VCSELs were also used to realize TDRC [62], experimentally solving [63] the Santa Fe task using a polarization-rotated feedback. VCSEL-based RC were further analyzed experimentally in [64].

Another promising category of TDRC systems are those based on passive photonics, offering the advantage of eliminating the need for external driving power to operate the reservoir. The first reported implementation [65] utilizes a nonlinear passive element; a semiconductor saturable–absorber mirror, showing good performance on multiple tasks. Shortly after, a high performance RC based on a coherently driven passive cavity was reported [66]. The cavity is based on a fiber loop and input/output couplers, as depicted in figure 3(d). Here, the nonlinearity is provided by the intensity detection at the photodetector after the linear mixing of optical fields through multiple round trips in a high quality factor cavity. The temporal evolution of a passive cavity's field E_{pass} can be generically described as follows:

$$E_{\text{pass}}(t) = \kappa_{\text{in}} E_{\text{in}}(t) + \kappa_{\text{fb}} E_{\text{pass}}(t - \tau_{\text{fb}})\, e^{i\phi}, \tag{11}$$

where ϕ is the roundtrip phase, E_{in} is the input field coupled with strength κ_{in}, and κ_{fb} is the feedback strength. In [66], this scheme yielded state-of-the-art performance on the NARMA-10 task. This was particularly interesting, as it showed that a linear optical reservoir with output photodetection can effectively solve demanding tasks. Using high enough optical power, the distributed Kerr nonlinearity in optical fibers can be exploited for TDRC, as was reported in [67]. In addition to task-independent evaluations, the study showed good performance on the Santa Fe task harnessing the Kerr nonlinearity in addition to the input layer and output layer nonlinearities (discussed more in section 2.4).

Recently, several implementations of TDRC based on PICs have been proposed and demonstrated. The primary appeal of these integrated approaches lies in their reduced form factor and potentially lower power consumption. This trend has been further fueled by significant advancements in the integrated photonics ecosystem, including ongoing progress in standardized fabrication processes and an increasing variety of foundry-accessible platforms such as silicon (SiPh), indium phosphide (InP), silicon nitride (SiN), TFLN, among others [68–74]. One of the challenges of integrated photonic TDRC is chip area, especially when considering the inclusion of longer delays. Longer delay lines enable larger time-multiplexed networks, but they are not feasible in integrated platforms where the losses are typically higher. This is the primary reason such implementations feature a lower upper limit on the number of nodes compared to fiber-optic implementations, where kilometers of fiber can be used with very low propagation losses.

The first reported on-chip photonic TDRC implementation combined a SL with a short external cavity for delayed feedback [75]. By using multiple delays and short θ, the number of nodes is increased from 6 to more than 300. This scheme was successfully tested on the Santa Fe time series prediction and the nonlinear channel equalization tasks. Shortly afterwards, a monolithically-integrated InP photonic circuit combining a laser and a 5.4 cm delay line was proposed and experimentally demonstrated [76, 77]. The system operated with 23 virtual nodes at 0.87 GSa s^{-1}, achieving performance comparable to implementations realized in benchtop setups. In this scheme, postprocessing techniques were also used to boost the number of nodes and the overall performance. Preprocessing and postprocessing are discussed in more detail in sections 3 and 4, respectively.

While earlier works proposed using networks of MRRs for photonic RC [78, 79], the first TDRC implementation based on SiPh MRRs was presented in [80], exploiting TPA and FCD for the nonlinearity. This implementation was able to solve the temporal XOR task and the Iris image classification. Due to the small size of the MRRs, these reservoirs exhibit a small MC. As illustrated in figure 3(e), this can be improved by including an external feedback loop, which has been done in follow-up numerical study [81]. The system was reported to be capable of solving tasks requiring longer memory, such as the NARMA-10 task, and later experimentally demonstrated on boolean tasks [82]. Extending the memory of nonlinear MRRs is also possible through cascading an array of linear MRRs, which was proposed in the numerical study [83]. An increase from MC = 2.4 in the single MRR case to MC $\approx$ 10 when 4 linear MRRs are added in series was reported. When 10 linear MRRs are considered, the obtained NMSE on the NARMA-10 task is slightly below the one obtained in [81], but slightly higher on the Mackey–Glass task. Although cascaded MRRs are challenging to fabricate owing to tight tolerances and sensitivity to process variations, tuning mechanisms can mitigate these limitations. MRRs were also numerically investigated in [84], where a dense grid of evanescently coupled nonlinear microcavities is considered for reduced chip footprint. The Mackey–Glass task and optical signal recovery task were solved, while elucidating the distinct impacts of TPA, FCD, and Kerr nonlinearity on the consistency and dimensionality of the reservoir.

The dynamics of MRRs were recently analyzed in depth in the context of RC [85]. In addition to MRRs, other resonators were investigated, including TDRCs based on integrated stadium cavities, as

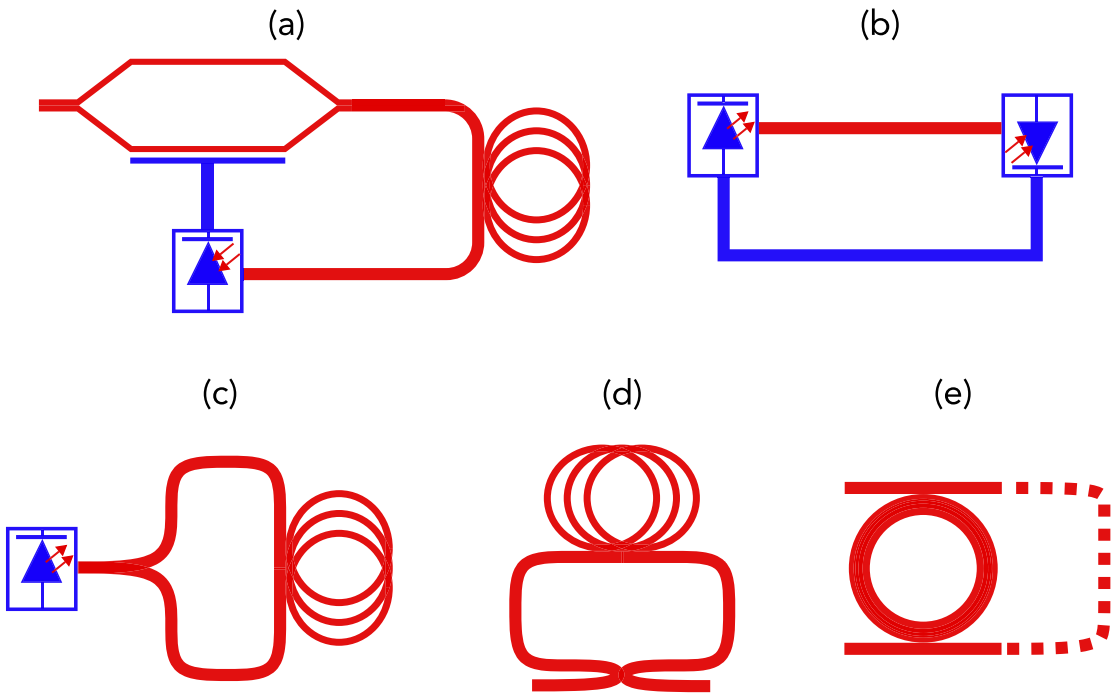


Figure 3. A few examples of the photonic TDRC schemes found in the literature. Examples of optoelectronic schemes include: (a) optoelectronic oscillator, (b) SL with electronic feedback. Examples of all-optical schemes include (c) SL with optical feedback, (d) optical loop (fiber-based or PIC-based), (e) PIC-based MRR with optional external feedback. Red (blue) color denotes optical (electrical) paths.

reported in [86, 87]. In [61], a PIC-based TDRC scheme was proposed showing good numerical results on some benchmark tasks while driven in the asynchronous regime. In this study, an MZI is used in feedback configuration, and a heating element reconfigures the dynamics. In this case, the clock cycle was set much longer than the feedback $T_{\text{clk}} \approx 3\tau_{\text{fb}}$.

2.3. Spatially-distributed photonic RC

In spatially distributed RC, the reservoir nodes and their connectivity are physically implemented. Each node is probed to yield the node state corresponding to the input clock cycle. Contrary to standard common-practice TDRC, the node states in spatially-distributed RC can be probed during the steady state portion of the response, although this is not a strict requirement. The weighted sum of the node states yields the reservoir's prediction, as it is the case in ESN software implementations. We report the advancements of spatially distributed RC in two sections: PIC and diffractive optics implementations.

2.3.1. PIC-based spatially-distributed RC

The first proposal of photonic RC dates back to 2008 [88]. In this numerical study, the use of nanophotonic coupled SOAs was motivated by its nonlinear relation between the input and output powers, which looks like an S-shape and resembles the upper half of the commonly used tanh activation function in RC. The benchmark task was signal pattern recognition and the baseline was a standard tanh ESN. This approach, as illustrated in figure 4(a), later developed into what was coined the 'swirl' topology [89], and ultimately removing the SOAs and relying only on a passive circuit of waveguides and splitters [90] in the first chip-based experimental demonstration of photonic RC. In this work, the responses were measured from 16 nodes (11 usable), and the temporal XOR was solved, along with optical header recognition and the isolated spoken digit recognition tasks. One particular challenge that this kind of spatially-distributed PIC-based schemes face is the need for long waveguides to slow down the reservoir response to be detectable by the readout electronics in the GHz range. The associated waveguide losses become non-negligible, and this introduces limits on scalability. In a more recent numerical study [91], an improvement of the passive swirl PIC reservoir (called the four port architecture [92]) was augmented with an SL in the feedback loop to enhance the computational capacity beyond that of the passive reservoir, by introducing nonlinearities at the reservoir layer. The performance was reported for the strong and weak data injection regimes for the Santa Fe task. The same topology was also explored numerically using nonlinear MRRs as nodes [79]. Smaller reservoirs were explored, such as the

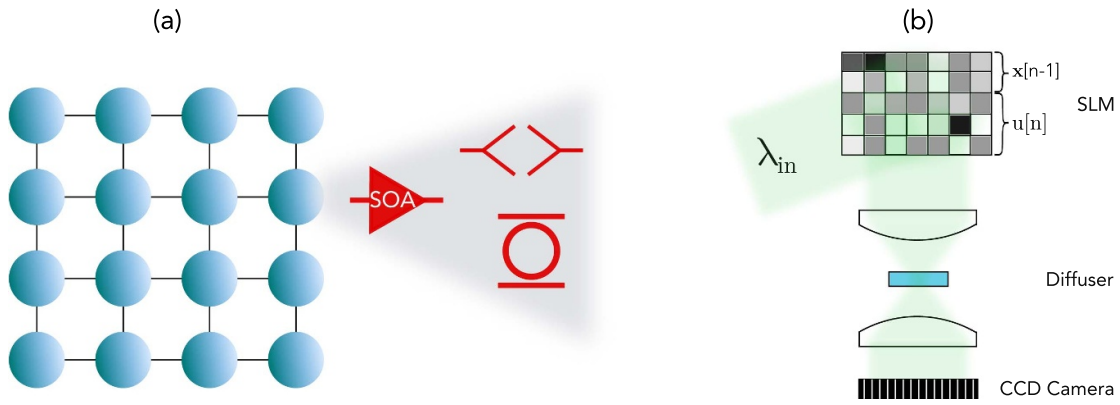


Figure 4. A couple of examples from the spatially distributed RC literature. In PIC-based implementations (a), various optical elements can be used as nodes, such as SOAs, passive splitters/combiners, and MRRs. In an example of a free-space implementation (b), a laser shines on an SLM, on which the current input and the past state are encoded. The light is reflected into a diffuser for random mixing, before detection at the CCD camera.

SiPh photonic crystal reservoir presented numerically in [93]. Inside the photonic crystal cavity, linear mixing results in a rich output response, which can be probed from different positions in the cavity via readout waveguides. There is a tradeoff between the number of readout waveguides and the quality factor of the cavity. In this case, a larger readout dimensionality would come at the cost of the reservoir's memory.

These kinds of scalability limitations motivated the exploration of what is sometimes termed *spatiotemporal RC* in PICs, combining both spatially-distributed and temporally multiplexed RC. In the spatiotemporal configuration of [94], the authors employ a silica-based, telecom-grade PLC to implement an integrated, scalable photonic TDRC scheme utilizing a coherent photonic processor [95–97]. While less compact than silicon photonics due to its lower index contrast, the PLC offers reduced propagation losses and improved thermal stability. An interferometric circuit splits the light into multiple spatial channels and applies preprocessing through controlled delays and MZI complex weighting before feeding the signals into several ring cavities in parallel. The output of each cavity is then detected by a high speed photodetector. This combination of spatial and temporal multiplexing yields a total of 512 nodes. It was tested on solving the Santa Fe time series and solved the MNIST hand digit recognition with 91.3% accuracy. Despite this value being far below moderate (≈95%) or the state of the art (99.8%), it represents a significant advancement in integrated PNNs in general. Spatiotemporal multiplexing can already be considered as a departure from the standard single, shallow reservoir layer. Such non-standard RC schemes will be discussed in section 7.

2.3.2. Diffraction-based spatially-distributed RC

Efforts towards free-space RC implementations aim to address the limitations of PIC-based spatially-distributed RC. Free-space implementations allow for a much larger number of physical nodes compared to on-chip approaches, but this advantage comes with the trade-off of a bulkier setup. In [98], a 5×5 array of VCSELs was used in a diffractive optic imaging setup. The setup allows the neighboring lasers to be coupled into each other, weighted primarily by a programmable SLM in the imaging plane. Under the assumption of the small angle approximation, the undiffracted light of one laser can be combined with the first diffracted order of the adjacent lasers. The output beam is then focused into a spot and detected by a CCD camera. Through injection locking of 8 lasers with an external laser, the nonlinearity was tuned to resemble some well-known activation functions used in software NNs [23]. This reconfigurable setup was later experimentally demonstrated for photonic RC [99], where it was shown to successfully solve the bitwise XOR task and the 3-bit header recognition task. It is worth noting that the control and stabilization of operation conditions of coupled laser schemes is a challenge; laser mismatches, differing losses, and noise from multiple sources introduce local instabilities that alter signal complexity and degrade global synchronization [100]. The use of imaging setups, such as the one depicted in figure 4(b), can be harnessed to scale up. In [101], an imaging setup comprises an SLM for the input intensity modulation and a diffractive optical element for the reservoir connections. The reservoir is essentially a recurrent network of Ikeda maps. A DMD implements the output weights, and the system is tested experimentally for 2025 nodes. In principle, the scheme can be scaled to tens of thousands of nodes.

Table 1. Key experimental milestones in standard photonic RC and reported performance. Benchmark tasks are discussed in detail in section 6.2.

Year	Milestone	Demonstrated tasks
2012	Optoelectronic TDRC [40]	TI-46 WER $= 4 \times 10^{-4}$, Santa-Fe NMSE $= 0.124$.
2012	All-optical TDRC [45]	Channel equalization SER $= 3 \times 10^{-4}$, IPIX NMSE $= 2 \times 10^{-3}$, TI-46 WER $= 0.03$.
2013	High-speed GSa s^{-1} photonic TDRC [46]	TI-46 WER $= 4 \times 10^{-4}$, Santa-Fe NMSE $= 0.1$.
2014	All-optical passive nonlinearity TDRC	Channel equalization SER $< 10^{-4}$, IPIX NMSE $= 2 \times 10^{-3}$, TI-46 WER $= 0.02$.
2014	On-chip spatially-distributed RC	One-bit delayed XOR BER $< 10^{-4}$, 5-bit header recognition.
2015	High-performance passive fiber-cavity TDRC [66]	NARMA-10 NMSE $= 0.048$, channel equalization SER $= 0$.
2018	On-chip TDRC [75]	Santa-Fe NMSE $= 0.1$, channel equalization SER $= 0.02$.
2019	Large-scale (16 000+ nodes) spatially-distributed RC	HAR 92%.
2021	Ultra-compact TDRC with a single MRR	One-bit delayed XOR BER$< 1.4 \times 10^{-3}$. Iris dataset 99.3%.

On the other hand, uncontrolled scattering can generate speckle, which has proven useful for computing. The idea is based on the observation that light scattering in complex media is random [102] and thus performs a random projection [103] of the incident optical field [104]. This means that, while the resulting signal may appear to be scrambled, the underlying structure of the information is still preserved. In [105], a complex medium, specifically a diffuser, is leveraged for nonlinear mixing. The reservoir's past state and input are encoded onto a phase-only liquid crystal SLM. Both the input and the reservoir state are recorded onto the SLM. The scheme solved the Mackey–Glass time series for multiple time steps up to the Lyapunov time. A similar speckle-based scheme realized 50 000 nodes in [106] to solve the KS chaotic time series. The nonlinearity is implemented through the phase encoding of the SLM at the input layer, and through the intensity detection by the camera at the readout layer. Furthermore, this scheme was recently implemented to realize an optical random NN [107] without the recurrence of RC, which was demonstrated on computer vision tasks. Speckle can also be induced in a MMF—and multimode waveguides in general—as they support multiple guided modes with differing phase velocities, which interfere after a sufficiently long distance. It was introduced for RC in [108], where it was reported to solve the Mackey–Glass time series and the Japanese vowels classification. A numerical investigation in the same study also suggested the possibility of implementing it on a SiPh chip with a 100 μm multimode waveguide. Speckle in MMF was also leveraged in [109], where the input is phase-modulated, and the speckle dynamics are tunable via the wavelength of the injected field. Thus, the nonlinearity is provided at the input. It was tested on the Santa Fe time series prediction, where good performance was achieved. Speckle-based photonic RC was studied numerically for multimode waveguide ring resonators [110]. Speckle was also used in [111] to mix the input before injecting it into a reservoir based on a LA-VCSEL. In this case, the complex transfer matrix of a MMF is multiplied by the Boolean-encoded input patterns, generated by a DMD, and displayed onto the LAVCSEL. Recurrent connectivity between photonic neurons arises from the LAVCSEL's intra-cavity field interactions and carrier diffusion. The readout layer is implemented by encoding (binary) trainable weights onto a second DMD, whose output is imaged onto a large-area detector to directly yield the network's computational result. The system was tested on the temporal XOR task, among others. Table 1 chronologically lists key early experimental milestones discussed in sections 2.2 and 2.3, which laid the groundwork for more recent developments reviewed in section 7.

2.4. Experimental considerations

The performance of a physical RC system depends not only on the underlying physical dynamics but also, inevitably, on the experimental setup. This makes it essential to dedicate effort to the setup design and characterization in order to accurately assess RC performance. In the following, we focus on common undesired effects, examine whether they degrade or enhance performance, and outline strategies used to address them.

Setup noise is a major limiting factor in photonic RC implementations, originating from various sources such as thermal and shot noise in photodetectors, noise from electronic and optical amplifiers, and quantization noise in ADCs, among others. While the overall effect of noise on RC performance is generally detrimental, the severity depends on the specific task. Nevertheless, some internal noise can be beneficial for training purposes, removing the need for regularization [112, 113], although this is typically implemented in software rather than in hardware experiments. It has also been shown numerically

(using an ESN), that some optimal amount of input noise can in fact enhance the short-term and long-term prediction accuracy of reservoirs [114]. Otherwise, in general, noise has been shown to reduce both MC [115] and IPC [39] of ESNs, primarily because it degrades the consistency property [116]. Similarly, a useful experimental consideration is to characterize RC performance as a function of the SNR, an approach adopted since the inception of ESNs [117] and since the earliest photonic demonstrations [35, 45, 65, 66]. An important reason for reporting performance as a function of the SNR of the experiment is that it enables meaningful comparisons between different implementations under similar output signal conditions. In [118], the Ikeda-based TDRC reservoir was studied under different SNR conditions, and it was shown that the robustness could be boosted by engineering the input preprocessing step. This will be discussed in more detail in section 3. More recently, numerical investigations also consider the performance under different amounts of noise, which is calculated based on various noise sources, such as the noise associated with the readout electronics [119]. Furthermore, ambient fluctuations are detrimental to the SNR. This necessitates proper thermal and mechanical stabilization of the setup. For example, in the fiber-based cavity described in [66], the spool was housed within a dedicated nested box enclosure. This enabled the excellent measurement of the total memory of the system (i.e. the sum of linear, quadratic, and cross memories), which closely approximated the readout dimension. On the other hand, the system presented in [108], which employed a long MMF and followed a fundamentally different spatially-distributed RC scheme compared to the time-multiplexed approach in [66], demonstrated stable operation for up to one hour without any reported stabilization measures. In the spatiotemporal scheme of [94], the reservoir response was stable for 3 h thanks to the thermal stability of the PLC.

Other nonlinearities in the practical setup, which are not initially considered in a RC scheme, can notably impact the performance. For example, in [45], a precompensation step was applied to mitigate the nonlinear response of the modulation stage (see equation (6)), enabling the authors to isolate the intrinsic nonlinearities of the reservoir from those introduced by the input MZM. A similar approach was adopted in [66]. Numerical investigations in [67] considered in detail the input MZM nonlinearity, and showed that nonlinearities in both the input and output layers can enhance performance on some tasks such as the Santa Fe time series prediction. Additional nonlinearities can also arise from input signal generation hardware, including AWGs and DACs. These distortions, which are in the form of bandwidth limitations, phase delays, amplitude clipping, or waveform asymmetry, alter the structure of the input before it reaches the reservoir. While these effects are not part of the reservoir's intended nonlinear transformation, they can significantly influence the overall system dynamics, particularly in tasks requiring precise input encoding. A clear example of this is shown in [120], where the authors use a SiPh MRR for TDRC and study the impact of input waveform distortion. The ideal synthesized waveforms are compared to those generated in the lab (affected by RF amplifiers, DAC bandwidth, etc) and show that these imperfections significantly degrade performance in nonlinear tasks such as XOR, while linear memory tasks (AND/OR) are less affected.

In addition to signal generation and detection, amplification is often necessary in photonic RC setups and can introduce further nonlinearities and signal distortion. These arise both in optical amplifiers, such as EDFAs and SOAs, and in high-speed electronic stages used in the readout chain, including TIAs and post-amplifiers. In optical amplifiers, gain saturation leads to amplitude compression, while amplified spontaneous emission adds broadband noise that distorts the signal temporally and spectrally. On the electronics side, bandwidth limitations, latency, and nonlinearities in the gain stages can alter waveform shape and dynamics, particularly when operating near the edges of the amplifier's dynamic range. Overall, these effects can modify the underlying structure of the reservoir states and interfere with, or influence, its intrinsic dynamics. They could also enhance the system performance through additional nonlinear contributions. Table 2 summarizes some of the popular implementations of photonic RC with a qualitative description in terms of speed, footprint, and the energy bottleneck pertaining to each scheme.

3. Input encoding and preprocessing

In contrast to digital computing, which is based on encoding high ('1') and low ('0') states, the method of input encoding in analog computing can significantly impact the performance. Physical RC is no exception; the input layer essentially defines how the dynamical system will be driven, and as such how it would map the inputs to the reservoir state representations. The focus of this section will be on the intentional steps performed, whether to achieve fundamental reservoir properties, or to improve the performance. On this matter, we begin by discussing the physical realizations of the input layer, and then

Table 2. Summary of representative photonic RC implementations. All comparisons are given within the context of photonic reservoirs.

Scheme	Implementation	Energy bottleneck	Footprint	Speed
Optoelectronic oscillator	TDRC, external fiber delay, nonlinearity at intensity or phase modulator.	O/E/O conversion and electronic feedback chain.	Fiber-based setup.	Limited only by auxiliary electronics bandwidth.
On-chip passive	Waveguide-based delays, loops in TDRC, nonlinearity at detection.	Chip insertion loss and waveguide losses.	Small–Moderate (chip-scale).	Limited only by auxiliary electronics bandwidth.
SL	Laser diode with external cavity (LK dynamics), nonlinearity from the gain medium and interference inside the cavity.	Laser bias and control.	Fiber-based setup.	Tens of GHz bandwidth.
VCSEL	Single TDRC or spatially coupled 2D VCSEL arrays.	Laser bias and control.	Bulky in spatially-distributed RC (free-space setup), fiber-based setup in TDRC.	Tens of GHz per emitter.
Interferometric meshes + microcavities	Programmable linear networks; nonlinearity at detection.	Phase shifter tuning (platform-dependent).	Moderate (chip-scale), scales quadratically for dense meshes.	Limited only by auxiliary electronics bandwidth.
SiPh MRRs	Single add-drop or coupled ring lattices, rich variety of MRR intrinsic nonlinearities.	Thermo-optic tuning and stabilization.	Very small (chip-scale).	Depends on used nonlinearity (FCD few GHz, thermo-optic 10 s of kHz), and photon lifetime.
Free-space diffractive	SLM/DMD-based encoding, multimode-fiber or diffuser scattering.	Consumption mainly in illumination and SLM.	Bulky (free-space setup).	Medium update rates (SLM/DMD-limited), but massive amount of nodes.

proceed with the input masking protocol, which is critical to the functioning of TDRCs. We also report on a variety of other preprocessing techniques.

The most common way to inject the input to a photonic reservoir is through intensity modulation, e.g. using an MZM. As mentioned in section 2.4, the MZM's sinusoidal nonlinearity (voltage to intensity) participates in the overall RC performance, even if it is only used for input injection. To isolate these effects, either a precompensation step is done [45], or it is tightly operated within a quasi-linear region [67]. Naturally, the modulation depth plays a role and depends on the modulation scheme. For example, in phase modulation, the RF power is distributed into multiple sidebands whose amplitudes follow Bessel functions of the modulation index, whereas amplitude modulation in the small-signal regime produces mainly the carrier and the first sidebands [121]. Other physical input layers can be realized; for example through phase modulation [42, 86, 122, 123], or modulating the injection current of a SL [46], among others. Generally, when encoding, the range is limited (e.g. 2π in phase encoding) in order to preserve input separability, which would otherwise be harmed from the periodicity of the modulator response. Interestingly, it was recently shown, for a spatially-distributed RC configuration, that going beyond the 2π range may improve expressivity and yield better performance on some tasks [124].

For TDRC schemes, an important preprocessing protocol is typically carried out. Consider figure 5, where a low-order delay system is perturbed by an input. In general, the system's order and characteristic timescale govern its output dynamics. Consider a stable dynamical system that exhibits consistency, the response is usually split into two parts: the transient and the steady state. The steady state is governed by the parameter choice of the defined system, and the input drive. On the one hand, the reservoir's 'memory' is predominantly governed by the hyperparameter space (physical properties), such as the feedback strength, feedback delay (in TDRC), and nonlinearity type (due to the memory-nonlinearity tradeoff [125]). On the other hand, the input masking protocol plays a critical role in

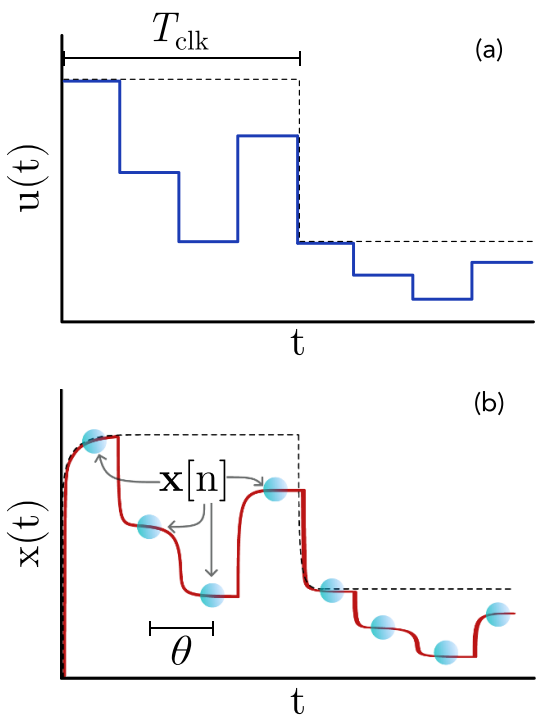


Figure 5. As an example, the input masking protocol is applied on a TDRC system with a single physical node, where $t_{nl} \ll \theta$ and $N = 4$. In (a), the input (dashed) and its masked version (blue) are shown. In (b), the response to the input (dashed) and the masked input (red) are shown. Sampled nodes are illustrated by the blue circles.

enhancing the computational expressivity of the system. It is applied to increase the effective dimensionality of the reservoir by breaking the symmetry among the time-multiplexed nodes, which would otherwise be correlated through receiving near-identical inputs. By setting N values within T_{clk}, the entire mask pattern is applied to each input to induce a diversity of node responses, thereby improving the separability of encoded input features. This is of course governed by the timescale of the individual node responses. The temporal distance between the masked nodes is thus given by:

$$\theta = \frac{T_{clk}}{N}. \tag{12}$$

The historical precedent has been to operate TDRC in the synchronized regime, i.e. $T_{clk} = \tau_{fb}$ [32]. However, many later studies consider the asynchronous regime, i.e. $T_{clk} \neq \tau_{fb}$. In standard practice, T_{clk} and N are chosen such that θ is shorter than the characteristic timescale of the nonlinear physical node ($\theta < t_{nl}$), thereby allowing adjacent nodes to be coupled through inertia. When t_{nl} is too small, the desynchronization by θ is typically employed, as discussed previously in section 2.2.

Fundamental to TDRC, the input masking protocol was introduced in the pioneering demonstration in [32], where different masks were constructed and tested on different tasks. In [118], masking strategies were explored numerically and experimentally within the context of an optoelectronic TDRC to improve the performance in the presence of noise. It was found that an i.i.d. mask consisting of 6 discrete levels yielded optimum performance, and further increasing the number of levels showed little to no improvement. In [126], the optimization of binary input masks was studied numerically with different motivations, considering a Mackey–Glass system as the nonlinear node. The first is to explore ways to construct optimized masks rather than employing heuristic approaches. The second is to maximize the utilized dimensionality of the system while using as few nodes as possible. Using fewer mask-driven nodes enables the shortening of the delay line (and consequently the input clock cycle), and thus enables faster processing speeds in the synchronous case. In [127], the impact of using a 'chaos' mask (recorded from a SL driven in the chaotic regime) is numerically studied and compared to other digital and analog masks (such as a variant of the six-level mask in [118]). The main insight was that a chaos mask induces a flatter distribution of node states. The chaos mask was subsequently leveraged to demonstrate superior performance on the Santa Fe chaotic laser dataset. A follow-up study augmented the findings with an experimental demonstration [128], where indeed the chaos mask outperformed the other digital or analog masks. However, no significant difference was found in the performance compared to a colored noise

mask for a given minimum cutoff frequency (≈6 GHz). In [94], a spatiotemporal complex-plane mask is applied through two phase shifters on each MZI, encoding both the amplitude and phase.

Since masks are usually prepared electronically, they require auxiliary (and often expensive) electronics, especially in high-speed setups. One way to tackle this is by looking at ways to generate the mask through simpler hardware or through leveraging inherent physical processes. This would bypass the need for digital generation, and would allow the input bandwidth to be dedicated to the input data itself. In [129], the mask is composed of two CW RF tones applied to a modulator. In practice, this can be realized with a couple of RF oscillators, which is substantially simpler than synthesizing an arbitrary input mask (e.g. with a high-speed AWG). The work in [130] proposes exploiting a SL with delayed optical feedback to generate the mask. By tuning the parameters of the system, different signal emissions were possible, based on repetitive quasi-sinusoidal patterns originating from limit-cycle dynamics of the system. The amplitudes of the comprising tones could be tuned to vary the resulting temporal pattern. It was applied on a generic tanh TDRC model and tested on the Santa Fe task. In [87], speckle patterns at the output of a MMF were exploited to perform the mask physically. Dynamic modulation (at 10 s of GHz) of the input phase alters the spatial profile of the emitted speckle pattern. While the mask itself is generated from a physical process in this case, it is worth noting that the (very fast) switching is itself controlled by auxiliary electronics, namely a deterministic RNG, possibly implemented through an FPGA. Another numerical study considers a spatial, passive photonic circuit, to perform the masking procedure [131]. The output can then be sent optically to the TDRC system (in this case, a SL-based TDRC). In another numerical study [119], the input masking protocol was replaced by a passive cavity reservoir's fast 'self-masking' dynamics. This was achieved by leveraging the interplay between T_{clk} and τ_{fb}, specifically when $\tau_{\mathrm{fb}} < T_{\mathrm{clk}}$.

In addition to improving expressivity, the input layer can be used to improve memory retention. One approach is through engineering the feedback loop, e.g. by injecting a filtered version of the feedback in an optoelectronic scheme [132]. Another approach is by feeding the past input in reservoirs with low intrinsic memories [75, 133]. Applying the past input on the input layer was studied numerically in [134], and recently experimentally demonstrated in [135]. Interestingly, in addition to enhanced performance on some benchmark tasks, the results show the possibility of bypassing hyperparameter optimization, and instead only tuning the injection strength of the past input. Alternatively, the input layer can be harnessed to bypass physical feedback altogether. In a recent work, the input signal is preprocessed to artificially incorporate temporal memory within a feed-forward, multilayer network constructed from DFB lasers [136].

The input encoding obviously influences the SNR of the entire setup. For example, in a numerical study [137], a multi-input strategy to a PIC-based spatially-distributed RC is explored, motivated by mitigating the effects of on-chip propagation losses between the physical nodes. In this setting, supplying multiple nodes on the chip with the same input was found to significantly improve the performance. In the scattering-based RC setup of [105], basket encoding is introduced to mitigate the loss of information caused by the binary modulation constraints of the DMD. Rather than mapping a scalar value to a single binary pixel, basket encoding represents it as an n_{bin}-bit binary code, where the active bit(s) correspond to a smoothly varying bin position. This structured, redundancy-enhancing mapping preserves similarity between nearby values in the Hamming space, thereby reducing the impact of bit-flips and better retaining the geometry of the original continuous state space.

In addition to input encoding, there may also be room for feature engineering. In some tasks, such as computer vision, a single input may consist of multiple features. Many such features are redundant and do not contain more useful information, and these can be safely discarded with image processing techniques. In [138], significantly higher classification accuracies were achieved using a combination of HOG and PCA, which reduced the number of components down to one fifth, while still explaining more than 90% of the variability in the data.

4. Readout and postprocessing

The readout is where the reservoir's node states are observed, often performed with an optoelectronic conversion in photonic RC. This prepares the signal to be processed, e.g. by a computer, to apply the learning method. The typical readout consists of a PD with associated amplification circuitry, followed by an ADC. Because the PD's detection bandwidth is far below the optical carrier frequency, it responds to the time averaged optical intensity over its bandwidth. Consider, for example, a simple phase-modulated coherent scheme consisting of a fiber loop and input/output couplers. The PD current I_{pd} is proportional to the optical intensity, which is governed by the interference of an arbitrary number

i of coherent fields $E_i(t)$, leading to the standard relation [139]:

$$I_{\mathrm{pd}}(t) \propto \sum_i |E_i(t)|^2 + 2\sum_{i<j} |E_i(t)||E_j(t)|\cos\left(\Delta\phi_{ij}(t)\right), \tag{13}$$

where the cross-terms depend on the relative phase $\Delta\phi_{ij}(t)$ between the interfering fields. After phase-dependent linear mixing, the resultant intensity variations are detected at the PD. Similarly, this can be looked at from the point of view of an all-pass loop power transfer function [140]. This detection nonlinearity is harnessed in several passive PIC-based implementations [61, 90, 94], where it plays a fundamental role in their operation. It can also be accessed in diffractive setups, typically employing an array of CCD pixels [141], where the intensity would be spatiotemporally averaged. Additionally, an inverted ReLU-like saturation-based nonlinearity can be exploited [106], which can be controlled by setting a longer exposure time or a stronger irradiance on the camera.

High-speed readout electronics can be costly. Hence, an analog readout was proposed for photonic TDRC [129, 142] to reduce postprocessing overheads. It was implemented using a MZM, on which the weights were encoded via an AWG, followed by a balanced PD and a RLC circuit to perform the summation. This chain carries out the inference, while a separate readout chain is utilized during the training phase. In the early versions of the spatial PIC-based swirl architecture [90], a high-speed PD and an ADC were required for each node (a total of 16), which introduced overheads in terms of energy consumption and latency. This motivated the study of an all-optical readout in [143], where each node's output is weighted using an optical modulator, which was then sent through a combiner tree structure before being detected at a single PD. The main goal of that study was to find a way to set the complex-valued weights at the readout when the individual states are no longer observable (discussed later in section 5). This all-optical readout was then investigated in simulation in [144, 145], highlighting the fact that the TIA saturation nonlinearity can be harnessed to improve the performance when a single PD is used.

These different methods of implementing the readout also required investigating the accuracy of the encoded output weights. This was studied in [146], addressing the problems of quantization and noise when applying the weights through a specific training method (discussed in section 5). Finally, this integrated all-optical readout was demonstrated experimentally in [147]. MRR-based SiPh weight banks [148, 149] have been investigated as an optical readout layer in [150]. Initially, the MRRs are set such that their resonances align with the optical carriers. The complex-valued weights are then encoded using a phase shifter on each MRR to adjust the amount of light coupled out, followed by another on the bus waveguide to adjust the phase. As such, through exploiting the spatial, temporal, and spectral domains, thousands of effective nodes can be realized in an integrated photonic implementation. It is worth noting that the study implements an unconventional RC approach, which will be discussed later in section 7. Besides, different kinds of PDs can be used depending on the target application. For example, a Kramers–Kronig coherent receiver [151] is explored in the context of a a 64-QAM signal equalization application [152].

In addition to the various implementations of the readout, a few postprocessing techniques have been explored. One area that could benefit from postprocessing is expanding the limited number of accessible nodes in an integrated TDRC implementation, where delay lines are size-limited. For example, in [75], the authors exploited multiple effective output delays from a single physical delay loop to increase the number of tapped nodes. Instead of introducing additional physical delay lines, the waveform circulating in the loop was sampled over an extended time window spanning multiple round-trips, such that the temporal evolution at delays $k\tau_{\mathrm{fb}}$ ($k \in \mathbb{N}$) was concatenated to form an enlarged reservoir state vector. This strategy effectively increases the dimensionality of the reservoir without modifying the photonic hardware, at the cost of a reduced T_{clk} which scales with $1/k$. In addition, the authors reduce θ by sampling at a higher frequency than the masked injection rate, thereby recording different values thanks to the continuous output transience within one mask duration. Similar types of postprocessing were tackled in another PIC-based TDRC implementation [76], which compared between the standard readout postprocessing, the halving of θ, and the doubling of τ_{fb}. However, contrary to [75], T_{clk} remains constant with respect to τ_{fb}. This means that the state matrix being constructed is effectively a concatenation of the response to the past input and the current input. It was shown that four times the number of nodes could be yielded by applying both techniques (what the authors refer to as double readout length), and that resulted in a significant accuracy boost on the Santa Fe task.

Multiple physical delay lines can also be used in the readout. In [153], multiple delay lines were used to improve the performance compared to a higher number of nodes with a single delay line. The idea of using multiple physical feedback loops was also explored on an optoelectronic RC in [154], motivated

by performance improvement on several benchmark tasks including NARMA-10, spoken digit recognition, and nonlinear channel equalization. The performance was also evaluated as a function of the system parameters, namely the difference between the two delays, the two offset phases, and the feedback strength. A double optical feedback was also implemented on a SL-based TDRC scheme to improve performance on simultaneous tasks [155].

The participation of the predicted output signal in the feedback loop was reported in [156, 157]. Instead of the internal states of the reservoir being in the feedback loop, the linear combination of the weighted states is performed by a high-speed FPGA and sent back into the reservoir. The weights are first trained offline before being deployed on the FPGA. With output feedback, the reservoir can run without external input by feeding its previous output back as the new input, creating a self-sustaining loop. This enables the autonomous generation of periodic signals (e.g. sine waves) or complex random patterns, as demonstrated in [157], which can be useful for high-speed signal generation applications. The system achieves high precision frequency generation error 7.5×10^{-5}, whereas random pattern generation is limited by experimental noise (measured standard deviation $\approx 2 \times 10^{-3}$ at the PD/ADC chain, while the time-multiplexed states have a standard deviation ≈ 0.2), highlighting the necessity for fast, low-noise output hardware for stable feedback operation.

5. Optimization and training

5.1. Hyperparameter optimization

While in standard RC theory the input and reservoir weights are fixed and untrained, in practice it is desirable to operate the system in a dynamical regime favorable for a particular task. Usually grid search methods are employed to scan the parameter space, but another approach is to use BPTT to optimize the reservoir hyperparameters [158]. This was done in [159, 160] to optimize the input and also learn the output weights on the optoelectronic TDRC scheme. Another approach is based on Bayesian optimization of the reservoir hyperparameters. It was proposed in [161], where it was numerically studied on TDRC and ESNs. It was then implemented in photonic RC in the context of a large-scale diffractive setup [162]. The experimental results on the HAR show a considerable 5% improvement (from $\sim 86\%$ to $\sim 91\%$), in addition to 30% gains in search time.

5.2. Training

Offline learning with regularized regression is by far the most widely reported in the literature, and has been discussed in section 2.1. Therefore, we focus on other learning methods that have also been explored and applied within the context of photonic RC. Popular alternatives include gradient-based techniques, such as BPTT. The analog readout proposed in [163] featured an online training mechanism, where an FPGA provides the input and output processing in real-time, computing the output weights using gradient descent. This is especially useful for real-time task processing, especially in telecom applications. The approach was applied on a nonlinear readout layer in a follow-up study [164]. In [101], a reinforcement learning flavored method is used to train the output weights encoded on a DMD. The weights are first randomly initialized, and a routine similar to stochastic gradient descent is applied for some iterations until minimization is accomplished. In [145], in the context of the proposed integrated all-optical readout, the regression would require a complex-valued target, even though post-detection it would be a real-valued quantity anyway. In this context, the authors use BPTT to avoid assuming an arbitrary value of the phase prior to detection, which would have greatly reduced the solution space in the complex regression. In [94], the limitations of a large input matrix in the MNIST data set motivated the training of the readout using gradient descent.

Evolutionary algorithms have also been explored. For example, in [143], a black box optimization technique called covariance matrix adaptation evolution strategy is adopted in the context of training an all-optical readout, where the individual reservoir states are no longer directly observed (they are combined at a single PD). In this case, the output weights applied through optical modulators still need to be learned despite the opacity of the individual states. It was shown to perform only slightly worse compared to the baseline case (complex regression with fully observable states).

The learning phase can also contribute to enhancing the robustness of physical RC systems. Sensitivity to environmental fluctuations, laser drift, input nonidealities, among other effects plague photonic analog systems. This was tackled in the multi-wavelength training strategy proposed in [165] and experimentally demonstrated on fiber nonlinear distortion compensation in [166]. This training technique relies on minimizing the MSE on multiple wavelength channels, enabling a twofold increase

in the stable operating range of the photonic chip. Questions of robustness were also addressed in [167] by proposing an adaptive model selection scheme based on reinforcement learning.

Another departure from standard learning methods is a study where emergent self-adaptation was realized without BPTT or manual setting of the output weights [168]. In that work, an integrated PNN of MRRs with PCMs exhibits emergent, backprop-free self-adaptation (volatile + non-volatile memory) and achieves 98.2% on MNIST using simple logistic-regression readouts and wavelength multiplexing.

One could argue that employing more involved optimization and training methods departs from the original spirit of the RC framework. Indeed, RC was initially attractive in software implementations due to its simplified training of a single output layer, in contrast to the more complex optimization required for conventional RNNs [169]. In hardware implementations, however, the RC paradigm is primarily defined by the physical realization of the reservoir itself, and less so by the optimization/learning algorithms employed. Historically, the difficulty of training full RNNs, primarily due to vanishing and exploding gradients, motivated the sidestepping approach embodied in the ESN framework. Although the LSTM architecture was introduced earlier [170] to tackle these challenges, its widespread adoption occurred later, driven by advances in digital compute resources and the emergence of gated recurrent units [171]. Notably, this period roughly coincided with the first photonic RC demonstrations. Conceptually, the canonical definition of RC, and its hardware implementations, do not strictly correspond to the same computational framework, as evidenced by the significant departures explored in physical realizations over the years.

6. Performance evaluation and applications

To systematically evaluate the performance of diverse physical RC implementations across various platforms and technologies, standardized benchmark tasks are employed to provide a controlled environment for comparative analysis. The evaluation methods are broadly classified into task-independent metrics and benchmark tasks. Task-independent metrics allow probing the intrinsic properties of the reservoir itself. On the other hand, benchmark tasks evaluate application-oriented performance. Benchmark tasks can be classified into a variety of task types, whether to imitate unknown dynamics, to predict the next step in a time series, or classification of data. Task-independent metrics provide insights on the trade-offs between memory and nonlinearity, and the overall behavior of the dynamical system. For reservoirs with tunable hyperparameters, task-independent metrics allow the choice of a suitable operation point. In the following subsections, we report some of the widely used RC metrics and benchmarks which have been applied to photonic RC. We also refer to a recent overview of RC benchmark tasks in [172].

6.1. Task-independent evaluation

Autonomous dynamics The most straightforward way to probe a reservoir is through observing its autonomous dynamics. This is done by setting the input to 0, and biasing the reservoir with an initial state. The long-term behavior, depending on the choice of hyperparameters, can be stable, oscillatory, or chaotic [173]. Bifurcation diagrams are used to show the steady state value(s) for a given choice of hyperparameter(s). A standard choice throughout the literature is to bias a stable reservoir close to a point of instability to generate highly 'expressive' dynamics. While this can be studied in active photonic implementations, it is not possible for passive implementations, where the steady state is always 0 in the absence of input and regardless of the choice of hyperparameters.

Lyapunov exponent The largest Lyapunov exponent of a dynamical system quantifies the average exponential rate at which two trajectories, which start from infinitesimally close initial states, separate/converge in the phase space [173]. Let $||\delta(t)||$ denote the Euclidean distance between the trajectories. For sufficiently small $||\delta(0)||$,

$$\|\delta(t)\| \approx e^{\lambda_L t}\|\delta(0)\|, \tag{14}$$

where λ_L is the largest Lyapunov exponent. If $\lambda_L > 0$, the distance grows with time (chaotic); if $\lambda_L < 0$, the distance decays with time (stable); and if $\lambda_L = 0$, the system is at the edge of stability. For discrete-time systems with unit time step $n \in \mathbb{N}$:

$$\|\delta_n\| \approx e^{\lambda_d n}\|\delta_0\|, \tag{15}$$

where λ_d is the discrete-time Lyapunov exponent. More generally, if a continuous-time system with exponent λ_L is sampled every Δt, then $\lambda_d = \lambda_L \Delta t$ and $\|\delta_n\| \approx e^{\lambda_L n \Delta t} \|\delta_0\|$. In RC, one is typically concerned with the largest conditional Lyapunov exponent along the input-driven trajectory [174]. A negative value ensures consistency and fading memory, whereas a large positive value destroys consistency under repeated inputs. Operating near the edge of stability ($\lambda \to 0^-$) often maximizes the combination of expressivity and memory.

Generalization test The generalization test [175] is performed by constructing M input sequences of length i, denoted $\{u_0^0, u_1^0, \ldots, u_{i-1}^0\}$ to $\{u_0^{M-1}, u_1^{M-1}, \ldots, u_{i-1}^{M-1}\}$, which are drawn from a uniform distribution. An identical sequence of length j, $\{v_0, v_1, \ldots, v_{j-1}\}$, is constructed and concatenated to each of the above sequences. The reservoir response to each input at the last time step are recorded in a matrix $\in \mathbb{R}^{N \times M}$. The choice of i and j is crucial for accurate assessment. For example, i is chosen large enough to ensure the reservoir reaches an input-driven steady state, before assessing convergence after j steps. The so-called GR, corresponding to the rank of the response matrix, is then computed, e.g. by using singular value decomposition. A thresholding of the singular value matrix is typically applied when a noisy system is considered.

Kernel test The kernel test [175] evaluates the reservoir's ability to separate different input sequences through its internal state representations. A number M of distinct input sequences of length T, denoted $\{u_0^0, u_1^0, \ldots, u_{T-1}^0\}$ to $\{u_0^{M-1}, u_1^{M-1}, \ldots, u_{T-1}^{M-1}\}$, are generated from a uniform distribution. Each sequence is independently fed into the reservoir and the response at the final time step to each input is recorded. All the responses are concatenated in a matrix $\in \mathbb{R}^{N \times M}$. Similar to the generalization test, the rank of this matrix is computed. A high KQR indicates that the reservoir produces well-separated responses to different inputs. The normalized difference between the two ranks is defined as the CA $=$ (KQR $-$ GR)$/N$.

Linear MC This test quantifies the ability of a reservoir to recall past inputs from its current state. It measures how accurately the reservoir can reproduce delayed versions of an input signal, for each discrete delay. Specifically, for an input sequence $u[n]$, the memory task at delay k aims to reconstruct $u[n-k]$ from the reservoir state $\mathbf{x}[n]$. MC was introduced in [115] and applied to the then-new framework of ESNs. The test is as follows:

(i) An input sequence $u[n]$ consisting of i.i.d. samples drawn from a uniform distribution over the interval $[-0.5, 0.5]$ drives the reservoir,

(ii) The output states are collected, and the states during the washout period are discarded,

(iii) The training targets are $u[n-k] \quad \forall k \in \mathbb{N}^+$ (theoretically) and the output weights are found for each target using linear regression.

It is given by:

$$\mathrm{MC}_k = \frac{\mathrm{Cov}^2(y_k[n], u[n-k])}{\mathrm{Var}(y_k[n])\,\mathrm{Var}(u[n-k])}, \tag{16}$$

$$\mathrm{MC} = \sum_{k=1}^{K} \mathrm{MC}_k, \tag{17}$$

where MC_k is the capacity to reconstruct an input delayed by k steps. Theoretically $K = \infty$, but in practice a sufficiently large number is considered. Usually a threshold is set, below which the associated MC_k term is considered insignificant and is set to 0. Jaeger showed that MC is upper bounded by the number of readout variables, and it was shown to reach this upper bound in the case of a linear ESN [115].
IPC First introduced in [39], the IPC extends the idea of MC to quantify the reservoir's ability to reconstruct both linear and nonlinear transformations of past inputs. These transformations of the past inputs constitute the targets, referred to as the basis functions. These target functions are required to be orthogonal, such that the IPC component measured on one does not influence the IPC component measured on the other. In theory, the total IPC is computed by considering an infinite number of basis functions, which means considering an infinite number of polynomial degrees and memory steps. In practice, only a finite length of the input can be used. Thus, a key assumption is made which permits the finding of relevant capacities; that for dynamical systems in general, these individual capacities decrease monotonically with increasing 'complexity' of the basis functions. Thus, a good search strategy for non-zero

capacities enumerates the basis functions by order of increasing complexity. A threshold is then applied to stop the search for capacities once the 'noise floor' is reached. This is discussed in detail in the supplementary material of [39]. The individual polynomials and their products form the target basis functions, which are constructed as follows:

$$y_{\{d_k\}} = \prod_k \mathcal{P}_{d_k}\left(u[n-k]\right), \tag{18}$$

where $\mathcal{P}_{d_k}$ represents the individual constituent polynomials of degree d and delay k, and u is the i.i.d. input. Depending on the chosen family of polynomials [176], u must be sampled from an interval where the target functions are orthogonal over that interval. The length of u is usually chosen to be sufficiently long (e.g. $\geq 10^5$) to reduce the capacities' noise floor. In addition to introducing the metric, the authors in [39] showed that the IPC is upper bounded by N.

6.2. Benchmark tasks

Temporal bitwise XOR This task was introduced in [177]. Contrary to other bitwise operations, e.g. AND/OR, the XOR is not linearly separable, i.e. it requires both memory and nonlinearity. The input is a binary stream, with $u[n] \in \{0,1\}$, and the target output is the XOR of two bits:

$$y[n] = u[n-k_1] \oplus u[n-k_2], \tag{19}$$

where $\oplus$ denotes the bitwise exclusive OR operation. When on-off keying is used, the accuracy is usually measured by the BER, given by:

$$\mathrm{BER} = \frac{1}{L}\sum_{i=1}^{L} \mathbf{1}(y_i \neq \hat{y}_i), \tag{20}$$

where y denotes the predicted bit and $\hat{y}$ the target. Under hard-thresholding, a fixed decision boundary is applied to assign each prediction to either class '0' or class '1'.

Santa Fe time series The Santa Fe benchmark is a univariate time series derived from laser-generated data recorded from a far infrared laser in a chaotic state for the Santa Fe competition [178]. The task is typically a one-step ahead prediction, but more steps could be also be considered for a more challenging version. Performance is typically measured by the NMSE, defined as:

$$\mathrm{NMSE} = \frac{1}{L\sigma_{\mathbf{Y}}^2}\sum_{\ell=1}^{L} \left\|\mathbf{Y}_\ell - \hat{\mathbf{Y}}_\ell\right\|^2 \tag{21}$$

where L is the total number of test samples, $\mathbf{Y}_\ell$ is the target output vector at training sample ℓ, and $\hat{\mathbf{Y}}_\ell$ is the predicted output vector at the same sample. The operator $\|\cdot\|$ denotes the Euclidean norm, summing the squared differences across all output nodes. The term $\sigma_{\mathbf{Y}}^2$ represents the variance of the target outputs over the entire test set and serves as a normalization factor. This normalization makes the NMSE a unitless metric, enabling direct comparison between tasks with different output scales. An NMSE of zero indicates a perfect prediction, while an NMSE of one corresponds to performance equivalent to always predicting the mean of the target outputs.

Mackey–Glass time series The Mackey–Glass time series is generated by a time-DDE exhibiting chaotic dynamics:

$$\frac{\mathrm{d}u(t)}{\mathrm{d}t} = \beta\frac{u(t-\tau)}{1+u(t-\tau)^{10}} - \gamma u(t), \tag{22}$$

with typical parameters $\beta = 0.2$, $\gamma = 0.1$, and $\tau = 17$. To generate the discretized time series, an integration step of 0.1 is commonly used, although sometimes a step of 0.01 is used. After discarding an initial transient, the task is to predict $u[n+k]$. As k becomes larger, the task becomes harder.

Kuramoto–Sivashinsky time series The KS equation is a nonlinear partial differential equation that models a variety of chaotic systems, such as flame fronts and thin film flows. It is typically written in the form

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x^2} + \frac{\partial^4 u}{\partial x^4} = 0, \tag{23}$$

where $u(x,t)$ is the scalar field. The task is to predict $u(t+\Delta t)$. This benchmark was introduced relatively recently for RC in [179].

Nonlinear channel equalization The nonlinear channel equalization task is a standard benchmark for RC. A sequence of transmitted symbols $d[n] \in \{-3,-1,+1,+3\}$ i.i.d. is constructed. The linear channel output is given by

$$\begin{aligned} q[n] = {} & 0.08\,d[n+2] - 0.12\,d[n+1] + d[n] + 0.18\,d[n-1] \\ & - 0.1\,d[n-2] + 0.091\,d[n-3] - 0.05\,d[n-4] \\ & + 0.04\,d[n-5] + 0.03\,d[n-6] + 0.01\,d[n-7], \end{aligned} \tag{24}$$

and is subsequently distorted by a nonlinear transformation:

$$u[n] = q[n] + 0.036\,q^2[n] - 0.011\,q^3[n] + \nu[n], \tag{25}$$

where $\nu[n]$ is an additive Gaussian noise, with its noise power determining the SNR. The task is to reconstruct the original transmitted sequence $d[n-2]$ when the reservoir is fed $u[n]$. Performance is typically evaluated through the SER, defined in a similar fashion to BER.

NARMA-k The (NARMA) task concerns the generation of a nonlinear function of the last k inputs:

$$y[n+1] = 0.3\,y[n] + 0.05\,y[n]\sum_{i=0}^{k-1} y[n-i] + 1.5\,u[n-k+1]\,u[n] + 0.1, \tag{26}$$

where k is the upper limit of discrete memory steps considered and $u[n] \sim$ i.i.d. $\mathcal{U}[0,0.5]$ is the input sequence to the reservoir. When k is large, e.g. $k=10$, this task is considered demanding as it combines both linear and nonlinear dependencies on input values deep in the past. The most common version uses $k=10$, although smaller k can be used when the intrinsic memory of the system is lower [119, 123].

Waveform classification The waveform classification task concerns the discrimination between different classes of temporal signal shapes, which may vary in amplitude, frequency content, or modulation pattern. The reservoir is driven by the sampled waveform, and a trained readout interprets this state to determine the class of the input waveform. Performance is typically quantified by the classification accuracy, defined as the ratio between the number of correctly classified waveforms and the total number of test waveforms.

6.3. Applications

A number of real-world applications are attracting attention in the photonic RC community. Below, we discuss several examples that have been explored in the literature. Here, we consider 'real-world applications' to mean tasks based on real datasets or problems with some industrial relevance.

Fiber transmission equalization Here the reservoir is tasked with reconstructing an original signal after it has been distorted by propagation through an optical link, typically including fiber dispersion and nonlinear Kerr effects, as well as other nonlinear sources along the channel. This task is particularly relevant to optical communications. Performance is typically evaluated in terms of NMSE or BER as a function of the optical link distance. It has been proposed for photonic RC and numerically simulated in [180], and experimentally validated in a follow up study [181]. Applications to telecom were also investigated in [182–184], and experimentally demonstrated in multiple studies including [166, 185, 186]. Overall, these works show that photonic RC can be applied within the real-world telecom industry. This is an important step, as the use of such processors would offload electronic DSP into the optical domain, paving the way for all-optical signal recovery approaches. Signal equalization was also demonstrated on PAM signals [187], and QAM signals [152, 188], where both amplitude and phase are modulated to maximize the available bandwidth. A detailed review of photonic RC for optical communication [189] summarizes the main developments and challenges in the context of telecom applications.

Optical header recognition This task addresses the problem of detecting and identifying short, predefined binary patterns at the beginning of optical data packets. These headers are used in optical communication systems for packet routing and synchronization. In this benchmark, the reservoir receives the

temporal waveform of the transmitted header, which may be degraded by dispersion, noise, or other transmission impairments. The response is trained at the readout layer to assign the correct header label, measured by SER. This task was tackled in numerous works in k-bit variants (for example 3 bits [99, 147] and 5 bits [90]), depending on the intrinsic memory of the system, which is required to implicitly store information about the last k bits of the input.

Computer vision Here, the reservoir is tasked with image identification. In photonic RC, spatial image data must first be converted to and encoded as a temporal sequence. The reservoir processes this temporal representation, mapping static/dynamic spatial patterns into high-dimensional dynamical trajectories. A trained readout layer then associates these trajectories with discrete output labels corresponding to image classes. Performance is typically reported as the classification accuracy, calculated as the percentage of correctly classified test images. Explored benchmarks within photonic RC include MNIST (handwritten digits) [87, 94, 168, 190, 191], Fashion-MNIST (apparel images) [87, 94], and similar datasets of grayscale images. To prepare this task for the temporal RC framework, additional preprocessing is needed as described in the separate approaches of [190] and [87]. The Iris flower classification was also featured in photonic RC [80]. In parallel to static image classification, RC has also been explored for HAR, a task that requires spatiotemporal classification of motion patterns. To make these datasets suitable for RC, raw video frames are either compressed into temporal feature vectors (usually preprocessed) or streamed sequentially to the reservoir, thereby preserving the temporal flow of the motions. As with static image classification, the readout layer is trained to associate the reservoir responses with discrete action classes (e.g. walking, running, or hand gestures). Performance is typically reported in terms of classification accuracy, with confusion matrices highlighting inter-class similarities, such as actions that feature some visual (temporal) resemblance. It was first experimentally demonstrated for RC in [138], and also featured in subsequent studies [192, 193], as well as recently in non-photonic RC implementations [194].

Spoken digit recognition The spoken digit recognition task (TI-46) concerns the classification of temporal audio patterns, specifically spoken digits (e.g. zero to nine). Input features are typically derived from MFCCs, which capture the spectral properties of audio signals in a manner aligned with human auditory perception, using e.g. the Lyon ear model. In this task, the reservoir processes the temporal sequences of audio features, transforming them into a high-dimensional dynamic state. A trained readout layer then maps this state to a discrete class label corresponding to the spoken digit. Classification accuracy serves as the primary performance metric. It has been featured since the earliest demonstrations [46] and continues to be a relevant task in the more recent literature [195].

IPIX radar The dataset from McMaster University [196] contains radar backscatter data from the ocean surface (sea clutter), recorded by the IPIX X-band radar. The task is to predict future clutter values $y[n+k]$, given an input of $y[n]$. Low and high sea states are included, corresponding to calm and rough sea conditions, respectively. It has been used in a multitude of photonic RC demonstrations.

Refractive index sensing Although less common, refractive index sensing can be performed in photonic RC, as described in [86]. By leveraging, e.g. a microcavity-based RC, detection of an external substance is possible through refractive index change. Changes in the refractive index modify the phase of reflected and transmitted light as well as the coupling efficiency to the detection probes, leading to measurable variations in the output intensities. These intensity signals are processed by a trained readout to estimate the refractive index change without requiring an explicit physical model of the sensing process.

Additionally, photonic RC shows strong potential in indirect processing applications. For instance, they can serve as preprocessors for deep NNs [197] or as PUFs [198], among other emerging use cases.

6.4. Multitasking photonic RC

Multitasking is another interesting avenue of applications for photonic RC. As the name implies, this is the case where a single physical reservoir is trained to solve multiple tasks simultaneously. The earliest suggestion of multitasking RC appears in [46], where a SL-based TDRC simultaneously performed isolated spoken digit recognition and speaker identification, by processing a single transient response of the system. The response indeed represented the features required for both tasks. Similarly, the bitwise operations (OR, XOR, XNOR, AND, NAND) were performed on the same input bitstream in [90]. Naturally, this gave rise to a question: can a single photonic reservoir solve simultaneously two tasks which depend on two different input streams? A numerical study addressed this in [199]. The proposed setup features a semiconductor ring laser emitting in 2 directional modes: clockwise and counterclockwise. This study

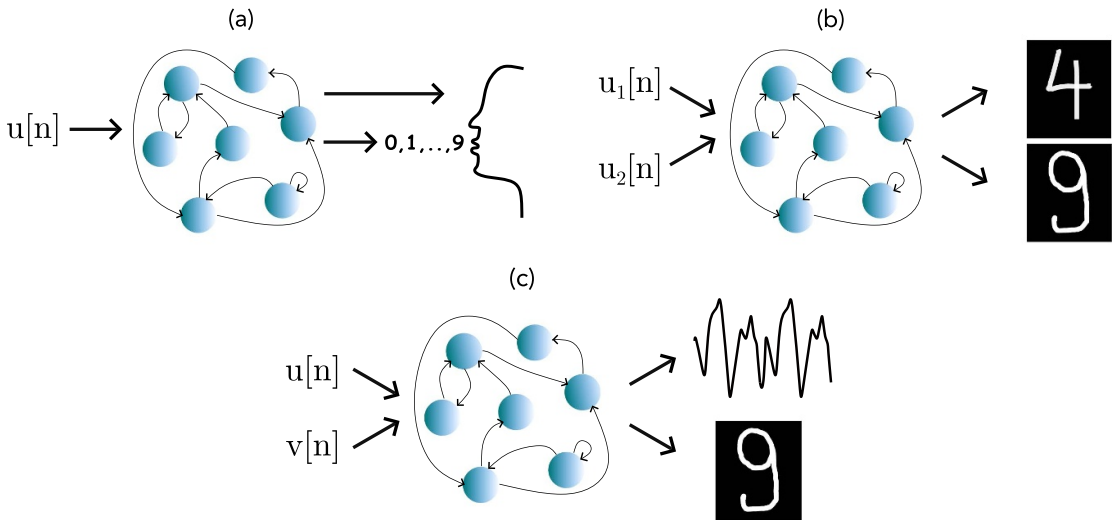


Figure 6. The different kinds of multitasking explored in the photonic RC literature: (a) homogeneous multitasking, (b) parallel processing, (c) heterogeneous multitasking.

highlighted that even under cross coupling conditions (i.e. the two modes are not isolated), multitasking can still be achieved. The two tasks investigated were the Santa Fe time series and the nonlinear channel equalization. In [200], the authors experimentally demonstrated multitasking by constructing several 'virtual reservoirs' (not to be confused with 'virtual nodes') on a single optoelectronic oscillator RC. This was done by temporally interleaving different masked inputs corresponding to the different tasks: nonlinear channel equalization, radar signal forecasting, and NARMA-10. The results were compared to the standard configuration for each task, showing in general that the nonlinear channel equalization and radar tasks preserved their performance. In addition, multiple instances of the same task (NARMA-10) were solved with an NMSE comparable to the standard optoelectronic RC approach [35]. It is worth noting that this particular virtualization strategy is harnessable when the considered nonlinearity timescale is 'instantaneous'.

Multitasking was also explored in a double feedback loop VCSEL-based TDRC scheme [155]. In the proposed setup, the two inputs are injected separately from one another. For the two feedback loops, the four possible combinations of using polarization-preserving and polarization-rotating configurations are explored. The two tasks considered were the Santa Fe timeseries and signal classification between sinusoidal and square signals. The best results were obtained when both feedback loops were in the polarization preserving configuration. In the experimental demonstration of [185], two different input streams were injected into the two polarization modes of a VCSEL under feedback. The results showed error rates of 0.3% and 3% for fiber distortions corresponding to 25 km and 50 km of transmission, respectively.

Speckle dynamics were also used to realize multitasking RC [109]. The experimental demonstration featured the simultaneous processing of multiple input streams of the nonlinear channel equalization task at 12.5 GSa s^{-1}. The scheme relies on two independent wavelength sources, which are each phase modulated with different input signals for the different tasks, before being combined through a coupler and sent down a long MMF. In this case, the same task (nonlinear channel equalization) was solved for two different inputs.

In [201], an equivalent MZM is implemented via a polarization modulator, polarizer, and a polarization controller. The inputs to the different tasks (NARMA-10 and IPIX radar) are encoded on the same polarization modulator by the same interleaving technique in [200]. The experimental results show an average NMSE = 0.21 on NARMA-10 and NMSE = 0.011 on IPIX. To yield the best performance, the bias of the readout layer is independently adjusted for each task. WDM can also be leveraged for multitasking, as was done in [94, 191, 202]. The tasks would be encoded onto different wavelengths of light, and the readout would typically feature a demultiplexer to separate the light into different respective channels for processing the different tasks.

One point to mention is that the degree and nature of multitasking is different across the discussed studies. We have observed in the photonic RC literature three distinct types of multitasking, which are illustrated in figure 6. We classify them as follows:

- *Homogeneous multitasking*: a single input instance is injected into the reservoir, and the response is mapped to multiple, distinct target outputs (e.g. simultaneous speaker identification and spoken digit recognition, or both XOR-2 and XOR-3 on the same input bitstream);
- *Parallel processing*: the same task is performed for multiple input streams (e.g. simultaneous channel equalization on separate input channels);
- *Heterogeneous multitasking*: multiple inputs from different native distributions and different target outputs (e.g. chaotic time series prediction vs nonlinear channel equalization).

In this way, the different types of multitasking can be clearly distinguished, and they can be independently and systematically evaluated as has already been done in e.g. [200, 202]. Finally, it is possible to allocate the entire usable bandwidth (for example the product of the number of WDM channels and the injection datarate) to a single task, potentially boosting the performance on that single task [202].

7. Unconventional photonic RC

What we have discussed so far may be described loosely as conventional photonic RC. These schemes comprise an input layer, a (single) reservoir layer, and an output layer, thereby resembling to an extent the classical ESN framework. Given the widespread adoption of the TDRC framework, we also classify it as conventional. In this section, we will consider some departures from these standard implementations in the recent years.

7.1. Supernetwork photonic RC

Within the RC community, systems composed of multiple reservoirs have been described using a variety of nomenclature. Sometimes the same term is used to describe different configurations, while in other cases different terms have been used to denote essentially the same concept. For example, the term *hybrid* RC [203] was used to refer to combining quantum and classical RC approaches, while previously in [204] the same term was used to refer to a combination of both *parallel* RC and *deep* RC on the same physical network. The term *hybrid* may not be required in the latter context, as the presented scheme in [204] seems to feature a deep architecture with full visibility of the hidden layers states, as described in [205]. The terms *delay-decoupled* RC [206, 207] and *parallel* RC have been used to essentially mean the same idea, while *delay-coupled* RC and *multiplexed networks* [208] were used to refer to multiple nonlinear nodes whose delays are coupled (the latter was later dropped in favor of the former in a follow-up study [209]). The terms *deep* RC and *serial* RC are sometimes used interchangeably [204, 210]. The term *serial* indeed implies a cascaded arrangement, but it does not inherently convey unidirectionality or the hierarchical abstraction that may emerge in deep RC.

To denote networks of networks (of nodes) in the context of RC, we suggest *supernetwork* (or *metanetwork*) as an umbrella term to describe the resultant network from any multi-reservoir configuration. Similar to standard RC, input, interconnectivity, and readout weight matrices would describe, generally, all the possible configurations of reservoir supernetworks. We suggest the distinction be made primarily between the use of a hierarchical supernetwork—already known in the literature as deep RC/deep ESN [211]—and a non-hierarchical one; a *lateral* supernetwork, as shown in figure 7. In standard deep RC, there is a unidirectional, layered chain of reservoirs: only the first reservoir is fed the input, and each subsequent reservoir/layer only takes as input the output of the previous adjacent layer. In lateral supernetworks, reservoirs typically share the same input layer, but may also exchange states depending on the connectivity. We thus use the term *parallel* RC (in line with the way it was used mostly in the literature) as those lateral RC networks where there is no cross coupling between their states (e.g. no cross-feedback in TDRC). Combining deep and lateral schemes would result in *deep-lateral* supernetworks. For supernetworks based on TDRC, the terms *delay-coupled* and *delay-decoupled* are used to distinguish whether lateral connectivity exists or not.

Constructing supernetworks is motivated by achieving richer output dynamics, or increasing the effective output dimensionality, or both. We have already briefly discussed in section 2.3.1 how spatiotemporal multiplexing techniques were used to expand the number of available output nodes within multiple PIC settings. The use of multiple reservoirs was proposed in [212] which was termed *Ensemble* RC. It was later applied to TDRC, considering two uncoupled sigmoid delay reservoirs in [206], where the authors noted an increase in MC when using two decoupled reservoirs vs. a single sigmoid with the same number of nodes. This was then further elaborated in [207], where the authors compare delay decoupled and delay coupled reservoirs with the standard case of a single node RC. The authors tested

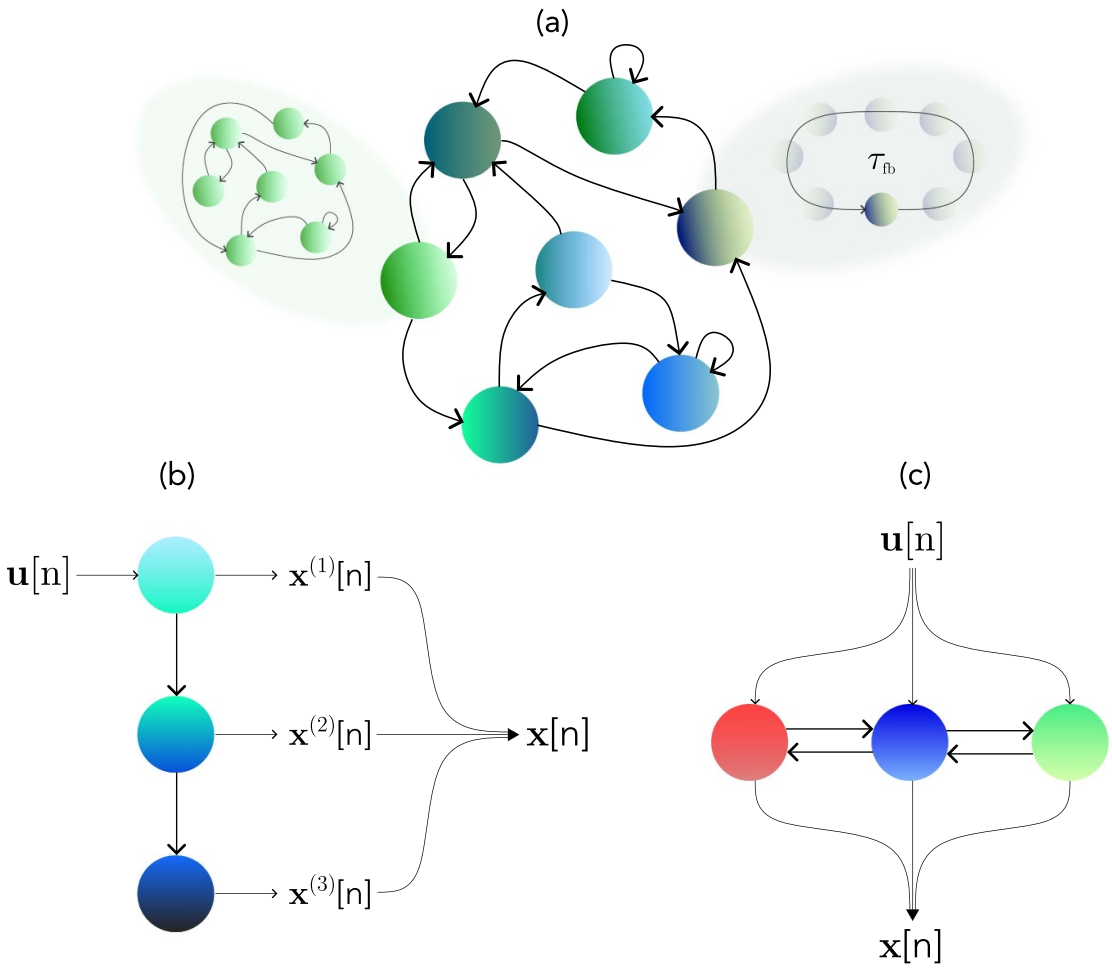


Figure 7. (a) Networks of individual reservoirs could be referred to as supernetworks. Depending on the inter-reservoir connectivity, different schemes can be realized, such as (b) deep RC or (c) lateral RC. To (conceptually) convert from a deep to a lateral configuration, at least one unidirectional link is instead made bidirectional, and the input is shared across the reservoirs. Combinations of both deep and lateral RC can be realized with at least three reservoirs.

for different numbers of reservoirs and with configurations of similar/different activation functions (by changing the sign on the exponent of the sigmoid). Larger number of reservoirs corresponded to consistently lower performance. In the case of decoupled reservoirs, mismatches between the reservoirs were found to improve the performance as it reduces the correlation between their responses.

Another extensive study on delay-coupled reservoirs was carried out with Stuart-Landau oscillators in [208], emulating coupled laser-based reservoirs. In agreement to [207], the authors also concluded that the performance degrades when a large number of physical nodes are used, whereas state of the art results were obtained with small-to-medium sized reservoirs. Mismatches were also considered and found to not degrade the performance. The results of these numerical studies showed that, in principle, such systems can be experimentally realized.

SL-based supernetworks were studied in [204], considering both parallel and deep connectivity. Two variants of deep topologies were considered, the first where the output layer was only probing the states of the second reservoir, and the second where both reservoirs' states were probed and combined. The latter showed superior performance on all tasks, likely exploiting the multiple timescales captured by both the 'shallow' and the 'deep' layers. Interestingly, the best performance was found with deep (parallel) RC for the Santa Fe (nonlinear channel equalization) task, showing that different tasks favor different supernetwork topologies.

The frequency-multiplexed fiber-based scheme presented in [213] was used to experimentally demonstrate a two layer deep RC in a follow-up study [195]. The states of all layers were combined together to form the output, where the inter-layer delays (from the physical connections) were compensated for by digital postprocessing. The deep scheme showed up to two orders of magnitude improvement over standard RC for two benchmark tasks. A 4-layer demonstration of deep RC was realized by cascading

four injection locked SLs [214]. It was applied to a fiber distortion correction task, where it outperformed a single-layer RC of the same type. However, the performance gains tended to saturate as the depth increased.

7.2. Next-generation photonic RC

In [215], the authors introduced NG-RC, which offers the removal of explicit recurrent dynamics while retaining standard RC functionality. This is done by constructing a feature vector directly from the input using time delay embedding and nonlinear mapping (e.g. by polynomial expansion), followed by standard regression to train the readout. This work showed that a linear reservoir combined with a linear readout applied to a nonlinear feature expansion (e.g. polynomial terms of delayed inputs) is equivalent to a NVAR model. The main advantages of NG-RC are its ability to achieve high performance with significantly shorter training datasets and a reduced hyperparameter space, owing to the elimination of the recurrent network and the use of direct nonlinear transformations of time-delay embedded input data. It is worth noting that both the NG-RC approach, and those photonic RC schemes which utilize a linear reservoir and an output nonlinearity, remove the nonlinearity from the feedback. NG-RC was first demonstrated in photonics in [216]. In this work, the photonic NG-RC leverages Rayleigh backscattering to realize a nonlinear mapping between input and output features through coherent interferometric mixing and a $|E|^2$ detection readout. This nonlinear projection delivers strong performance across benchmark tasks while retaining the main benefits of digital NG-RCs such as a short warm-up time, efficient training, and simple hyperparameter tuning. The authors also propose a masking strategy that emulates different dynamical behaviors (unit, fading, or random), and show that, for different tasks, the performance depends on the choice of mask. Performance comparable to software-based NG-RC was achieved on the Lorenz task, and state of the art performance was achieved on the KS task. A PIC NG-RC implementation was reported shortly after in [150], with an experimentally demonstrated processing speed of 60 GHz. The entirely integrated scheme relies on an on-chip modulator which injects the data into an array of delay lines. Linear mixing between the delayed inputs takes place in a star coupler. The output is then split into multiple channels for processing (and applying the $|E|^2$ nonlinearity). The Santa Fe and the Lorenz tasks were successfully solved. In addition to its compactness, the approach leveraged WDM to employ nine output ports, achieving comparable performance on the COVID classification task to that obtained with a single wavelength and 45 output ports. The nonlinearity is implemented at the readout. More recent implementations [217, 218] show that it is currently an area of growing interest, even applying it to quantum RC systems [219]. The key strength of physical NG-RC implementations lies in their feedforward-like architectures, which largely avoid the speed limitations inherent to delay-based reservoirs and recurrent dynamics more generally. It is, however, up to debate whether NG-RC complies with the standard definitions of RC that we briefly revisited in section 2.1, or whether it is a related but different computing paradigm.

7.3. Spiking-based photonic RC

Another avenue of unconventional photonic RC is combining it with another computational framework, such as SNNs. While SNN and RC are typically different computing paradigms, they have been both implemented (separately) using VCSELs. VCSEL spiking behavior was experimentally demonstrated in [220], showing that a VCSEL neuron produces a brief ($\sim$100 ps) spike only when sufficient subthreshold inputs arrived within a short temporal window. As such, a spiking RC based on VCSELs was first proposed and demonstrated in [221]. The authors experimentally demonstrate that, by combining the excitable spiking dynamics of the VCSEL with the RC/ELM framework, they could realize a neuromorphic photonic processor that computes using optical spikes. The system employs time-multiplexing of the spiking output to create time-multiplexed nodes, which are interconnected through the device's nonlinear integrate-and-fire and refractory dynamics. It was demonstrated on the Iris flower classification task, solving it with 97% accuracy, and thus highlighting the strong computational potential of VCSEL-based spiking RC. In a follow-up study, it was successfully demonstrated on the Mackey–Glass time series [222]. The use of optical spikes in the demonstrated VCSEL-based photonic SNNs offers some advantages over traditional non-spiking PNNs. Similar to standard RC, only the output layer is trained; however, spiking photonic RC additionally enables efficient operation with sparse signaling and sparse weight matrices, supports the use of simple binary weights, and maintains reliable performance even with relatively small training datasets. Continuous progress is being made in this direction with multiple recent studies using different platforms, such as using a fiber-based setup employing a single DFB laser with an integrated saturable absorber [223], and using integrated MRRs [224]. In [224], the authors demonstrate deterministic spiking dynamics with a single optical pulsing injection applied to a single MRR that integrates and fires to WDM signals, achieving 92% on the Iris classification task. MRR

self-pulsing dynamics [225] have also been explored for photonic RC. The interplay of TPA, FCD, and thermo-optic timescales was exploited to yield self-pulsing dynamics from a single nonlinear MRR [226]. This is implemented in an 8×8 array of low-Q factor SiPh MRRs. This approach, dubbed *non-fading memory RC*, is useful for example in sensing applications, where the extended memory of the system enables the matching of much slower input timescales.

7.4. Quantum photonic RC

Quantum photonic RC is an emerging direction that seeks to combine the strengths of photonic hardware with the non-classical properties of quantum systems. By leveraging quantum states of light, such approaches aim to enhance the IPC beyond classical limits, for instance through quantum parallelism or entanglement. Currently the field is very much a work in progress. While promising, quantum photonic reservoirs face the typical challenges of quantum systems, making this an active area of ongoing research [227–230]. An in-depth discussion of quantum photonic RC is outside the scope of this work.

7.5. Stochastic RC

Tied to quantum RC, stochastic NNs have been demonstrated using probabilistic neuron activations in an optical network implementation [231]. Stochastic RC has been theoretically investigated in [232], where certain classes of stochastic ESNs were shown to form universal approximating classes. The key advantage is that the effective number of computational nodes can scale exponentially with hardware size by using probability distributions of outcomes as readouts, enabling compact devices to realize large functional state spaces. It was shown that when noise is small, stochastic RCs can outperform deterministic reservoirs of similar complexity in certain tasks. The main drawback, however, is the need to estimate the probabilities with sufficient precision (a large number of runs would be required).

8. Conclusion

8.1. Summary and discussion

In this review, we have grouped the progress in photonic RC literature by theme. We have considered both pioneering demonstrations from the earlier literature, as well as the latest advancements in the recent literature for a comprehensive report on the state of the art of photonic RC. From the discussion in sections 3 and 4, we saw that the input and output layers, in addition to the reservoir itself, also influence the RC properties of photonic systems. We reviewed the optimization and learning algorithms used across different schemes (section 5), highlighted common benchmark tasks and surveyed relevant real-world applications (section 6). We also proposed a set of terminologies aimed at unifying certain concepts in the literature, especially concerning multitasking RC and reservoir networks comprising subreservoirs. Finally, we turned to recent trends, with particular attention to developments in unconventional RC, including e.g. spiking or quantum approaches (section 7).

One can view physical RC as a gateway to the broader field of physical computing, owing to the relatively simple training requirements. It allows researchers to concentrate on the physical implementations, and to gain insights from computation with dynamical systems through testing on a variety of physical substrates [22]. Photonics provides a rich platform for RC, owing to the many degrees of freedom of light, for example temporal, spectral, spatial, and polarization, among others. The flexibility of realizing RC schemes with fiber-based/free-space/integrated setups has enabled a wide range of implementations. Overall, the RC paradigm [18] and the time-multiplexed equivalence [32] have contributed to the revived interest in using photonics for computing.

One of the primary challenges for photonic RC is hardware scalability, particularly with respect to reproducible operation at the level required for mass production. Architectures that rely on nonlinear transient responses-such as time-delay reservoirs-are especially vulnerable when the underlying optical nonlinearities are difficult to control. By contrast, photonic VMM accelerators operate within a linear and well-defined regime, making their reproducibility considerably higher than that of systems based on uncontrolled nonlinear dynamics. Although fabrication tolerances such as phase errors, waveguide losses, and coupling imperfections introduce deviations, these are systematic in nature and can typically be corrected through calibration, for instance via thermal or electro-optic tuning of phase shifters. Once calibrated, the system remains stable on mature photonic integration platforms. Of course, such systems are still plagued by significantly fewer elements (compared to standard GPU), noise and effective precision limits, among other challenges. This distinction partially explains the growing popularity of photonic RC implementations that employ linear optical systems combined with nonlinear detection despite the inherent functional advantages that a nonlinear optical system can offer for computing. We provide in

Table 3. Comparison between photonic hardware accelerators and RC.

	Strengths	Limitations
VMM/MAC accelerators	• Extremely fast and energy-efficient for large-scale, dense linear algebra • High throughput, low latency • Scales with parallelism	• Addresses linear tasks • No inherent memory
RC	• Built-in memory • Harnessable nonlinear dynamics • Low-cost, rapid training • Addresses nonlinear tasks	• Efficiency and scalability are not straightforward • Performance governed by reservoir dynamics

table 3 a brief comparison between photonic hardware accelerators and photonic RC. Scaling photonic RC systems in terms of the node count is equally important for transitioning from proof-of-concept demonstrations to practical, high-performance applications. As summarized in table 4, different multiplexing strategies-time, wavelength, spatial, or their combinations-offer distinct advantages and trade-offs between implementation complexity, latency, and achievable node count. Practical design choices depend on the target performance and system constraints. When the goal is to maximize throughput with a moderate number of nodes, wavelength-multiplexed architectures with parallel readout are most suitable, which can make use of high-quality optical frequency comb sources [233] and low-crosstalk filters to preserve channel fidelity. If large virtual node counts are required and higher latency is acceptable, then time-multiplexing offers a simple and cost-effective way to scale the reservoir. Alternatively, spatially distributed networks based on devices such as DMDs and SLMs can achieve tens of thousands of nodes in a single pass, making them attractive for very high-dimensional processing, although they require careful control of optical alignment, stability, and noise, in addition to engineering the fading memory. Finally, for real-time, low-latency operation with many physical nodes, integrated photonic circuits that combine multiple degrees of freedom (e.g. time, wavelength, and space) provide a promising route, albeit with stringent demands on loss management and readout electronics.

Photonic RC has demonstrated competitive performance on many standard RC benchmark tasks, while leveraging the THz bandwidths of energy-efficient optical systems. From an applications point of view, photonic RC has already displayed competitive performance on application-specific computing, especially in the context of telecom applications. Continued advancements in the integrated photonics community are expected to fuel advancements in photonic RC and photonic computing in general, as PICs become scalable [234], flexibly programmable [235], efficiently designed [236, 237], and easier to characterize and control [238]. Indeed, recent progress towards photonic accelerators already display remarkable technological maturity [239, 240], harnessing dense integration for the tackling of VLSAI applications.

8.2. Perspectives and open challenges

Integrated platforms hold large promise for compact, stable, and manufacturable reservoir computers, yet they remain constrained by propagation losses, limited nonlinearities, and the chip area required for delay lines or large spatial networks. Future work should prioritize low-loss waveguide technologies, heterogeneous integration, and on-chip amplification schemes that avoid excessive power consumption. Improving these aspects will enable reservoirs with orders of magnitude more effective nodes, bringing photonic RC closer to task-relevant scales for telecom, RF, and high-bandwidth signal processing. While substantial optical power is still required to drive modulators, lasers, or nonlinear elements, this burden can be reduced through low-power modulators (e.g. TFLN, electro-absorption), passive cavity designs, resonant nonlinearities, and the development of standardized energy metrics analogous to energy-per-MAC in electronic accelerators. Such metrics would enable quantitative comparisons across architectures and help identify energy-optimal operating regimes.

A central challenge remains the development of strong, controllable, and fabrication-tolerant nonlinearities. Many optical nonlinearities are either too weak (Kerr), too slow (carrier effects), or too sensitive to fabrication variations (microcavities). Programmable or reconfigurable nonlinear elements, hybrid electro-optic nonlinearities, and designs exploiting naturally strong effects (e.g. semiconductor gain saturation or free-carrier absorption) could substantially broaden the computational regimes available to photonic RC. In parallel, most current implementations still rely on heuristically chosen parameters, masks, and dynamical regimes. Physics-aware machine learning frameworks that jointly optimize input encoding, masking, reservoir parameters, and post-processing offer a promising direction, enabling

Table 4. Comparison of different scaling strategies for photonic RC, highlighting how each approach scales, its main benefits, and its key trade-offs. We also provide one example for each approach.

Approach	How it scales	Advantages	Limitations	References
Temporal networks (TDRC)	Large number of nodes from a single nonlinear element, at the cost of latency.	• Very simple hardware • Compact footprint	• Small node count in high-speed settings • SNR degradation for long delays	[35]
WDM	Parallel channels: near-linear throughput scaling with number of channels.	• High parallelism • Comb sources can be used • Low latency	• Crosstalk between channels • Complex filters • Coherent phase control • Detector array often required.	[203]
Spatial networks (integrated)	High node density per area using e.g. compact MRRs.	• Low latency • Compact integration • On-chip processing	• Fabrication variability • On-chip loss • Thermal tuning and control overhead	[79]
Spatial networks (free-space)	Many spatial modes: massive node count.	• Very large state spaces from passive media • Minimal active hardware	• Crosstalk between modes • Stability issues • Designed fading memory (not intrinsic)	[106]
Deep RC	Richer dynamics through stacked reservoirs, without necessarily increasing node count.	• Increased expressivity (hierarchical dynamics) • Performance boost on hard tasks	• Accumulated noise • Careful tuning of inter-layer dynamics • More hardware overhead	[196]
Spatiotemporal-spectral multiplexing	Combines multiple multiplexing axes: potentially very large node counts and throughput.	• Best theoretical scaling	• High implementation complexity • Combined limitations of each axis	[94]

high-performance configurations that may be unintuitive or difficult to reach manually. Furthermore, this should be supported by the capability of modeling complex systems through building computationally efficient and precise device models for system-level simulations that go beyond the photonic layer, extending to the electronic and software layer. Such modeling should also consider environmental and aging aspects as well as specific use case frameworks that may set constraints on the computing workflow.

Much of the existing photonic RC literature continues to focus on classical benchmark tasks rather than domain-specific applications. Identifying the application niches where photonic RC genuinely outperforms digital processors-such as ultra-fast equalization, real-time classification of optical communication signals, event-driven sensing, RF signal intelligence, or adaptive control for photonic systems-is essential. Application-driven constraints (e.g. latency, bandwidth, or memory depth) can guide architectural choices, and these constraints may differ significantly across use-cases. Beyond classical approaches, spiking photonic reservoirs, quantum-inspired reservoirs, and architectures exploiting spatial, spectral, and temporal degrees of freedom simultaneously may unlock new computational regimes. Hybrid electronic-photonic reservoirs, or cascaded systems combining photonic front-ends with low-power electronic post-processing, also represent promising near-term opportunities.

A key objective is leveraging the full potential of optical physics while scaling RC systems to handle more complex and demanding tasks. This includes increasing the complexity and scale of photonic implementations, developing training algorithms that can cope with these more involved systems, and ultimately migrating such architectures to integrated (and potentially hybrid) platforms suitable for

volume production in photonic foundries. Strengthening collaboration with real-world application partners will also be crucial for identifying where the distinctive advantages of photonic systems can be most effectively employed. Relevant opportunities include native optical I/O application scenarios, protocols for effective time-multiplexing to exploit more of the temporal bandwidth of optical systems, and integrated platforms incorporating advanced functionalities such as non-volatile elements for reduced energy consumption or optical gain for improved nonlinearity and scalability.

In terms of broader impact, a recent study [241] showed that MC computed on anatomical brain connectomes decreases with age and relates to cognitive and neuroimaging markers, extending RC-based metrics to biological computation. General-purpose computing has been associated primarily with digital architectures, rooted in the formal mathematical framework established by Turing [242], and focused largely on idealized models of rational, structured thought. Yet this paradigm represents only a narrow subset of the computational processes that occur in natural systems. More recently, there has been interest in formalizing computation from a more general, bottom-up (physics-first) perspective [243, 244]. Within this context, continued advances in photonic and other physical realizations of RC hold the potential to further elucidate how information is transformed, stored, and processed in physical dynamical systems. We therefore view photonic RC as not only a promising avenue for practical applications, but also as a means to help deepen our understanding of the fundamental nature of computation itself.

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Data availability statement

No new data were created or analysed in this study.

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